

REVIEW

Open Access



Revolutionizing cancer treatment: engineering mesenchymal stem cell-derived small extracellular vesicles

Ameneh Shokati¹, Mahshid Akhavan Rahnama^{2*}, Ladan Jalali³, Saghar Hoseinzadeh⁴, Saman Masoudifar⁵ and Mohammad Ahmadvand^{2*}

Abstract

In recent years, use of exosomes is one of the important approaches to treat cancer. Mesenchymal stem cells are multipotent stem cells which are very putative sources for modulating the immune system and importantly inhibiting immune responses against tumor cells. Small extracellular vesicles (small EVs) are extracted from mesenchymal stem cells can have therapeutic approaches for delivering targeted drugs to cancer cells and also tumors via regulation of signaling pathways.

There are challenges in use of exosomes in cancer treatment which need to be improved. Engineering in small EVs can recover the role of exosomes such as drug delivery and therapeutic tools. There are preclinical and clinical studies in the field of mesenchymal stem cells derived from small EVs which demonstrates efficacy of these approaches in the field of cancer.

In consideration of problems in large scale production and so on, engineered small EVs as treatment approaches for cancers, it is resulted that they retain most characteristics of cells which are derived from, are responsible for their immunomodulatory impacts driving positive changes in the tumor microenvironment.

Keywords Mesenchymal stem cells, Engineered extracellular vesicles, Cancer

Introduction

Cancer is a global life-threatening issue and the second leading cause of death worldwide [1]. The challenge lies in developing effective therapeutic strategies due to cancer cells originate from the body itself, which evades immune responses and allows tumor growth. Hence, a major concept in cancer therapy lies in the enhancement of the immune system to produce a strong immune response to the cancer cells. In the 1970s, cell shedding lead to the discovery of extracellular vehicles (EVs), which were further characterized into distinct classes [2, 3]. EVs can be classified into three main subtypes according to their origin, namely Microvesicles (200–1000 nm), Exosomes or small EVs (30–150 nm), and apoptotic vesicles (800–2000 nm) [1]. Small EVs, small particles with a membrane

*Correspondence:

Mahshid Akhavan Rahnama
m-akhavanrahnama@farabi.tums.ac.ir
Mohammad Ahmadvand
mahmadvand@sina.tums.ac.ir

¹Department of Applied Cell Sciences, School of Advanced Technologies in Medicine, Tehran University of Medical Sciences, Tehran, Iran

²Cell Therapy and Hematopoietic Stem Cell Transplantation Research Center, Research Institute for Oncology, Hematology and Cell Therapy, Tehran University of Medical Sciences, Tehran, Iran

³Department of Biochemistry, School of Medicine, Urmia University of Medical Sciences, Urmia, Iran

⁴Department of Medical Biotechnology, Faculty of Advanced Medical Sciences, Tabriz University of Medical Sciences, Tabriz, Iran

⁵Blood Transfusion Research Center, High Institute for Research and Education in Transfusion Medicine, Tehran, Iran



© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

composed of a phospholipid bilayer, are characterized by their contents reflecting their parent cells and containing various biomolecules, including proteins, lipids, metabolites, and nucleic acids [4, 5]. The formation is a multi-step process that involves several critical stages. The process begins with the indentation of the cell plasma membrane, leading to the creation of early-sorting endosomes (ESE). As ESE progresses, late-sorting endosomes (LSEs) are generated, and multi-vesicular bodies (MVBs) are formed, containing a significant number of intraluminal vesicles (ILVs). These MVBs are then released as in recent years, substantial advances have been made in exosomes -based immunotherapy, demonstrating it as a viable and safe approach for inducing innate and adaptive immune responses. The discovery of MSC-derived small extracellular vesicles (MSC-small EVs) as a cell-free therapeutic material presents new opportunities for cancer treatment through exocytosis [6–8]. MSC-small EVs are generated and released by various cell types, including nervous cells, immune cells, tumor cells, fibroblasts, adipocytes, and epithelial cells, and bodily fluids such as saliva, amniotic fluid, breast milk, urine, malignant ascites, and plasma [9–12]. Small EVs have gained significant interest in the use of modified and engineered sEVs. These modifications aim to improve treatment quality and specificity while addressing potential limitations of natural sEVs. They play wide range roles in physiological and pathological processes [13], including modulating the immune system, cell death [14], inhibiting immune responses against tumor cells. They also play a role in cancer growth, metastasis, and angiogenesis [15]. Small EVs can also have antitumor effects [16]. They can be isolated and detected using techniques like differential centrifugation, precipitation, flushing separation, ultrafiltration, immune-magnetic bead method, microfluidic separation, and mass spectrometry [17]. Following extraction, the separated particles need to be characterized using methodologies like dynamic light scattering, transmission electron microscopy, nanoparticle tracking analysis [18].

MSC-derived small extracellular vesicles (MSC-Small EVs)

Mesenchymal stem cells (MSCs) are a class of multipotent stem cells that can be extracted from various sources, such as bone marrow, adipose tissue, umbilical cord blood, placenta, brain, lung, dental pulp [15]. Unique characteristics of MSCs have generally made these cells useful tools in the fields of regenerative medicine and treatment of malignancies by containing and delivering active biomolecules to the afflicted cells and tissues, including peptides, proteins, and RNA species. Like others, MSC- small EVs also contain lipids, proteins, and nucleic acids. Since these cells can be harvested from

various origins, MSC- small EVs from each tissue exhibit unique markers in addition to some common markers found across all sources ones [19]. Low immunogenicity, high biocompatibility, and sufficient stability are characteristics that have made MSC-sEVs preferable [20].

Bone marrow-MSCs (BMMSC) - small EVs are characterized by the presence of specific markers, including CD9, CD63, CD81, TSG101, miR-301, and miR-22. These MSC- sEVs have been shown to suppress the production of interferon-gamma by blood cells. Furthermore, they possess immunomodulatory properties and have been demonstrated to promote tissue repair, stimulate growth and proliferation, and exhibit anti-inflammatory effects [21, 22].

HUCMSC- small EVs characterized by the presence of markers including CD44, CD90, CD105, HLA-I, and CD29, in addition to microRNAs such as miR-26b and miR-206. These exosomes exhibit anti-inflammatory properties by reducing the secretion of inflammatory cytokines, particularly interleukin-6 and they also have the capacity to enhance angiogenesis [23–25].

Adipose-derived stem cells (Ad-SCs) possess a relatively stable membrane, allowing for prolonged storage. These MSC- small EVs are characterized by the presence of diagnostic markers such as CD105, CD90, CD73, CD63, CD44, CD29, CD13, CD10, CD9, MHC I, ALIX, and TSG101. Furthermore, these exosomes play a significant role in various biological processes, including tissue repair, immune modulation, and the inhibition of apoptosis and Decrease both visceral and cutaneous fibrosis [26–28].

Exosomes obtained from MSCs originating from various sources possess a range of contents that can be utilized in the treatment of diseases, particularly cancers. These exosomes can leverage their unique components for therapeutic purposes and serve as vehicles for delivering targeted drugs to cancer cells within tumor tissues.

One of the key molecules found in MSC- small EVs is miR-187. Research indicates that miR-187 can diminish the expression of the CD276 (B7-H3) molecule, which plays a role in activating the JAK2-STAT3 signaling pathway and is notably prevalent in tumor tissues and contributes to the evasion of immune responses by cytotoxic T cells and natural killer cells. A study conducted by Chuangui Li and colleagues demonstrated that BMSC-small EVs with overexpressing miR-187 inhibit the malignant characteristics of prostate cancer cells, suggesting their potential as a promising therapeutic option for this type of cancer [29].

In a separate investigation carried out by Shu-Cheng Liu and colleagues, it was demonstrated through in vitro and animal research that exosomal lncRNA PTENP1 derived from BMMSC, when introduced to bladder

cancer cells, inhibits the cancer phenotype via established molecular mechanisms [30].

Researchers have discovered in various studies that human umbilical cord MSCs (hUC-MSCs) promote apoptosis in ovarian cancer cells [31] and suppress the aggressive traits of malignant cells in lung cancer [32].

Lou et al. conducted a study using an animal model, which revealed that employing small EVs derived from Ad-SCs as a delivery system for miR-199a-3p in hepatocellular carcinoma enhanced the sensitivity of cancerous cells to chemotherapeutic agents that target the mTOR signaling pathway [33].

The findings from these studies indicate that the application of small EVs, particularly those derived from MSCs, in the management of cancers, notably solid tumors, has garnered significant interest. Continuous efforts are being made to enhance their efficacy and address current limitations.

An essential consideration is that the utilization of natural small EVs presents challenges, such as the potential for accumulation in non-target tissues, which can diminish their therapeutic efficacy. Additionally, the heterogeneous composition of these particles may complicate the effectiveness of the intended therapeutic target. Furthermore, when small EVs serve as carriers for pharmaceuticals, there may be concerns regarding the safety of the therapeutic approach. To address these issues and harness the unique capabilities and properties of small EVs for disease treatment in a precise and controlled manner, it is necessary to modify them, implement desired alterations, and subsequently employ them as therapeutic agents. The following outlines the various methods employed in the engineering of small EVs.

Strategies for engineering MSC- small EVs as drug delivery

Biological modification

Biological modification of MSC- small EVs is a crucial strategy for enhancing their efficacy as drug delivery systems. This approach involves altering the parent MSCs to produce exosomes with tailored therapeutic properties. Key aspects of biological modification include:

Genetic engineering

Genetic modification of MSCs enables for the overexpression of specific proteins and growth factors within their small EVs. Techniques such as viral transfection and CRISPR/Cas9 can be utilized to introduce genes encoding proteins, peptides, cytokines, and therapeutic agents that facilitate tissue regeneration. For example, MSCs can be engineered to generate small EVs that are rich in anti-cancer cytokines, providing substantial benefits for conditions like tumorigenesis and malignant tumors [2].

Ma et al. devised an endogenous loading method by pre-transfecting plasmids containing target genes into Ad-MSC to generate therapeutic small EVs that enhance angiogenic and osteogenic regeneration [3].

Furthermore, Huang et al. increased the levels of Pigment Epithelium-Derived Factor (PEDF), a protein with multiple functions known for its neuroprotective, anti-oxidative, and anti-inflammatory effects, in Ad-MSCs through overexpression. This modification aimed to assess its protective role in brain injury via both in-vitro and in-vivo experiments. The findings indicated that MSC-small EVs modified with PEDF effectively mitigated cerebral ischemia-reperfusion injury by modulating cell apoptosis and autophagy [4].

Zhang et al. genetically engineered MSCs to generate small EVs overexpressing miR-29a-3p. Treatment with engineered small EVs significantly inhibited migration and vasculogenic mimicry (VM) formation in glioma cells. Additionally, the anti-glioma effects of miR-29a-3p and miR-29a-3p-overexpressing small EVs were validated in vivo. This study highlights the potential of using MSCs to produce miR-29a-3p-enriched exosomes, which hold promise for anti-VM therapy and could serve as a complementary approach to anti-angiogenic treatments in clinical settings.

Additionally, small EVs isolated from MSC transfected with miRNA-124a demonstrated the ability to target glioblastomas (GBMs) and inhibit tumor growth in mouse models, leading to improved survival rates [5].

Exosomes, known for their stability and biosafety, are emerging as promising drug carriers for nucleic acid therapy. MSCs naturally home to tumor, inflammation, and injury sites due to surface tropism factors like CXCR4, which binds to SDF-1 on tumors. In a study, MSCs were engineered to produce small EVs with high CXCR4 expression, enabling targeted delivery. These small EVs were loaded with si-Survivin through electroporation, creating a novel genes-drug delivery system (CXCR4^{high} small EVs /si-Survivin). Experiments showed that this system effectively accumulates at cancer sites, releases siRNA into cancer cell, and knocks down the Survivin gene, preventing tumor growth.

Protein engineering

Direct protein engineering can significantly enhance the functional properties of MSCs-small EVs. By altering the surface proteins of small EVs, scientists can improve their targeting abilities. This process involves attaching antibodies or ligands that specifically interact with target cells' receptors, thereby increasing the precision of exosomes-mediated drug delivery [6]. One study demonstrated that modifying the small EVs' surface proteins can increase their targeting efficiency toward cancer cells. For example, researchers engineered MSCs-small EVs to

express the HER2 peptide, which improved their binding affinity to HER2-positive breast tumor cells. The article also suggests that utilizing MSCs ensures a sufficient supply of small EVs for research purposes [7].

Another approach involves targeting molecules that are highly expressed in cancer cells. For instance, hyaluronic acid (HA), which is prevalent in certain malignancies, can effectively target these cancer cells. In a study by Vogus et al., Hyaluronidase (HYAL) was linked to MSCs-small EVs containing doxorubicin and gemcitabine, resulting in the formation of MSC-Exo-HYAL. The authors reported that MSC-Exo-HYAL specifically targeted triple-negative breast tumor cells and significantly inhibited cancer growth [8].

Recombinant transmembrane proteins of small EVs represent a prevalent engineering method for modifying their surface properties. One example is “lysosomal-associated membrane protein 2b (Lamp2b)”, which acts as a surface protein of small EVs with a signal peptide [9]. Another strategy involves engineering peptide-Lamp2b fusion protein that incorporates glycosylation motifs at various sites. This glycosylation-stabilized peptide enhances the targeting capability of small EVs to cancer cells, such as neuroblastoma cells. Through genetically engineering, the proteins of interest can be combined with Lamp2b on the small EVs surface, serving as a specific target. The presence of homing peptides or ligands displayed on the external small EVs surface further increases the efficiency and specificity of their delivery [9].

Recently, Wang et al. utilized phage display techniques to identify a novel sequence of peptide, CSTSMLKAC (IMTP), which targets regions of cardiac ischemia. They subsequently modified MSC using lentivirus packaging technology, fusing IMTP with Lamp2b and incorporating it into the surface of MSC-small EVs (IMTP-small EVs) [9]. In mouse models of myocardial infarction, IMTP-small EVs notably enhanced the targeting of cardiac cells and reduced cardiomyocyte apoptosis [10].

In addition to enhancing targeting capabilities, genetically engineered modifications of Exomembranes can facilitate the attachment of therapeutic molecules [11]. A 2021 study demonstrated the creation of a recombinant expression vector for a cells membrane-penetrating peptide (CPP) fused with TNF- α (CTNF- α), which was successfully introduced into mesenchymal stem cells (CTNF- α -MSCs). Following this, TNF- α -anchored small EVs were isolated from CTNF- α -MSCs and exhibited notable anticancer activity [12].

Enhancing cargo loading

Biological modifications can also aim to enhance the loading capacity of small EVs for therapeutic agents. By altering the metabolic pathways of MSCs, researchers

can improve the internalization of nucleic acids or drugs into sEVs. This may involve optimizing the production conditions to ensure a higher quantity of therapeutic cargo is encapsulated within the resulting small EVs [6].

MSCs-EV can inhibit T-cells activity by delivering soluble factors and modulating signaling pathways. Research has demonstrated that the CD30 cargo in MSCs-EV enhances immunosuppressive effects by promoting the accumulation of adenosine [13].

Additionally, MSCs-small EVs have been demonstrated to promote angiogenesis, likely via the AKT/eNOS pathway, by enhancing the expression of miRNA-221-3p. These results indicate that MSCs-small EVs carries cargo that regulates angiogenesis in recipient cells [14].

Exos containing miRNA-222/223 have been linked to drug resistance in breast cancer. Specifically, the miRNA-23b cargo in MSCs-small EVs can trigger resistance to the proteasome inhibitor docetaxel in breast cancer cells [15].

Furthermore, MSCs- small EVs can induce resistance to fluorouracil in gastric cancer cells by enhancing mRNA levels of MDR, LRP, and MRP via the activation of Raf/MEK/ERK and calcium-calmodulin dependent protein kinase (CaM-K) signaling pathway [16].

Pan et al. utilized lentivirus to transfect miR-132-3p mimics as an angiogenic miRNA, into MSC, successfully isolating MSC-small EVs that overexpress miR-132-3p. In vitro studies showed that these small EVs can deliver miR-132-3p to target cells more effectively than natural small EVs, leading to reduced reactive oxygen species (ROS) generation, decreased apoptosis, and improved tight junction integrity in endothelial cells subjected to hypoxia/reoxygenation damage [17].

Similarly, Shen et al. discovered that miR-133b-modified MSC-small EVs in rats with cerebral- hemorrhage mediated neuroprotective effects through RhoA and ERK1/2/CREB signaling pathways, preventing apoptosis. Beyond protecting neurons, these small EVs also facilitate the transfer of miR-133b to astrocyte cells, modulating genes expression and promoting neurite- remodeling, thereby enhancing functional recovery after stroke [18].

Besides miRNAs, the incorporation of endogenous proteins, such as transcription factors, cytokines, and, enzymes have further enhanced the therapeutic efficacy of MSC-small EVs in various diseases. Wei et al. employed an adenoviral transfection system to introduce human anti-protease alpha-1 antitrypsin (hAAT) into MSC. They found that hAAT-MSC-Exos exhibited significant immunoregulatory benefits [19].

Improving stability and biodistribution

Enhancing the stability of MSC-small EVs can significantly boost their effectiveness as drug delivery systems. This enhancement may involve modifying the lipid

composition of the exosomal membrane to increase resistance to degradation in biological environments. Furthermore, engineered small EVs can be optimized for prolonged circulation in the bloodstream, which improves their delivery to target tissues [20]. One strategy involves incorporating glycosyl phosphatidylinositol (GPI) onto the small EVs surface. GPI serves as an anchoring structures for functional ligands, such as antibodies and RNAs, effectively protecting surface proteins of small EVs from hydrolytic degradation by proteases and guiding small EVs to cancer cells [21, 22].

Immunological modification

Immunological modification of MSC-small EVs represents a promising strategy to enhance their effectiveness as drug delivery vehicles, especially for targeting immune-related disorders and improving therapeutic outcomes in tumor treatment. This approach involves altering the immune characteristics of MSC-small EVs to modulate immune responses and boost their therapeutic efficacy. Key components of immunological modification include:

Surface modification with immune checkpoint molecules

By engineering MSC to express immune checkpoint inhibitors on the surface of their small EVs, scientists can enhance the ability of these small EVs to evade immune detection. This modification allows small EVs to remain in circulation longer and improves their accumulation at targeted regions, particularly in cancer microenvironments that immune evasion is crucial. Many tumors exhibit surface CD47, which interacts with signal-regulating protein α (SIRP α) on phagocyte cells, thereby reducing macrophage-mediated phagocytosis of tumor cells. Researchers have developed a small EVs -based immunal checkpoint blocker that antagonizes the CD47–SIRP α interaction [75, 76]. Azide-modified small EVs were conjugated with antibodies (dibenzocyclooctyne-derivatized SIRP α). These chemically modified small EVs were targeted to disrupt the CD47–SIRP α checkpoint on the cancer cells' surface, leading to a reduction in tumor cell engulfment by immune cells.

Incorporation of immunomodulatory factors

MSC-small EVs can be loaded with immunomodulatory agents, such as small molecules or cytokines, to increase their therapeutic effects. For instance, small EVs can be engineered to carry interleukin-10 (IL-10) or transforming growth factor-beta (TGF- β), both of which can inhibit inflammatory responses and improve tolerance in disorders [6].

One of the most studied cytokine that activate the therapeutic potential of MSC- small EVs is TNF- α [11, 23]. Exposure to TNF- α has been shown to enhance the

therapeutic efficacy of MSC-EV for various disorders. Mechanistically, mild induction of TNF- α shifts MSC toward a more anti-inflammatory phenotypes, leading to the release of a greater number of anti-inflammatory exosomes [11].

Another inflammatory mediator that boosts the regenerative capacity of MSC-small EVs is IL-1 β . Recent research indicates that IL-1 β treatment enhances the anti-inflammatory effects of MSC- sEVs in astrocyte cells through the NRF2 signaling cascade [11, 24].

Furthermore, IL-1 β -pretreated MSC- small EVs demonstrated improved anti-inflammatory effects in models of sepsis and osteoarthritis [11]. In addition to inflammatory cytokine, Yang et al. showed that interferon-gamma (IFN- γ) also enhances the capacity of MSC-small EVs to inhibit Th17 cells differentiation and alleviate colitis [25].

Targeting immune cells

Immunological modification can also enhance the ability of MSC-small EVs to specifically target immune cell. By decorating the surfaces of small EVs with ligands that bind to receptors on immune cell—such as dendritic or T cells—scientists can improve selective uptake of small EVs by immune cells. This strategy is particularly beneficial for directing treatment agents to modulate immune responses in chronic inflammatory and cancer conditions [20].

Additionally, umbilical cord- MSC-exosome can deliver miRNA-182, which increases tumor cell death by enhancing the proliferation of natural killer (NK) and T cells, as well as by regulating the sensitivity of tumor cells to immune attacks [83]. Zhou et al. engineered MSC-small EVs with galectin-9 siRNAs and oxaliplatin, exposing them to immune cells. The results demonstrated that these MSC-small EVs enhanced antitumor immunity by suppressing regulatory T cells (Treg) downregulation, promoting cytotoxic T cells enrollment, and altering macrophage polarization [26].

Tsai et al. developed two membrane-localized vectors (FGL1-TM vectors and Programmed cell Death-Ligand 1 (PD-L1) vectors) and co-transferred them into MSC (FP-MSC) using a bioengineering strategy. The resulting FP-MSC-small EVs expressed high levels of PD-L1 and FGL1 while retaining the immunomodulatory properties of unmodified MSC-small EVs. These engineered dual targeted MSC-small EVs showed potential in alleviating transplant rejection. It was confirmed that PD-L1 and FGL1 on small EVs specifically bound to their cell receptors, LAG-3 and PD-1, on their target cells. Importantly, FP EVs significantly inhibited T cells activation and proliferation in vitro as well as in a heart allograft model. Furthermore, FP EVs encapsulated with low-doses FK506 exhibited enhanced effects on inhibiting T cells proliferation, reducing CD8+ T cells density and cytokines

production in the heart grafts and spleens, inducing T reg cells in lymph nodes, and extending graft survival. Collectively, these dual-targeting EVs have the potential to synergistically enhance immune inhibitory signaling and slow transplant rejection [27].

Engineering for antigen presentation

MSC-small EVs can be modified to present viral antigens or tumor associated antigen (TAA), thereby enhancing their ability to stimulate an immune reaction against infected or tumor cells. This strategy aims to leverage the immune system to more effectively recognize and eliminate malignant cells [20].

Additionally, Folate (FA) is overexpressed in various cancers, including pancreatic cancer, whereas its expression in normal cells and tissues remains relatively low. Consequently, FA serves as an effective targeting ligand for the delivery of antitumor drugs to cancer cells [28]. Yu et al. proposed that hUC-MSCs-small EVs labeled with Folic acid (FA) and loaded with erastin can target triple-negative breast cancer cell, significantly slowing tumor growth [29]. Additionally, genetically engineered (small EVs -PH20-FA) incorporating FA can specifically target tumor tissue, enhancing the delivery efficiency of anticancer drugs [30].

Hyaluronic acid (HA) is often overexpressed in some malignancies. Hyaluronidase (HYAL), an enzyme that breaks down HA, can be utilized to improve tumor targeting [31]. Hence, HYAL and HA interaction can be utilized to improve the targeting of small EVs to tumors. For instance, a delivery system using HYAL-modified MSC-small EVs loaded with gemcitabine and doxorubicin effectively targeted triple-negative breast cancer, reducing tumor growth [8]. Similarly, metalloproteinases (MMPs) show promise in tumor targeting, as demonstrated by SiO₂ nanoparticles carrying MMPs substrate peptides that selectively target MMP-rich cancer cells. Other enzymes, such as cathepsins and heparinase, also hold potential for targeted cancer therapies [32].

Physical modification

This approach focuses on altering the physical characteristics of exosomes to enhance their stability, targeting capabilities, and overall therapeutic efficacy. Key strategies for physical modification include:

Surface coating and functionalization

Surface modifications can significantly influence how MSCs-small EVs interact with target cells. Coating MSCs-small EVs with biocompatible materials or attaching ligands can improve their specificity and effectiveness in targeting desired cells. For example, incorporating polyethylene glycol (PEG) can enhance circulation time and decrease immune clearances, while specific peptides

or antibodies facilitate targeted delivery to tumor cells [33].

Harting et al. induced the secretion of anti-inflammatory bone marrow MSC (BMMSC)-small EVs under stimulation with TNF- α and IFN- γ to precisely target sites of inflammation [34]. Although the surface markers and size distribution of MSCs-small EVs derived from naïve and inflammation-treated MSC were similar, differences were noted in their RNA, cytokines content, and protein composition. Notably, there was a further reduction in the pro-inflammatory cytokines' secretion, partly due to the COX2/PEG2 expression compared to naïve MSCs-small EVs [35].

Furthermore, small EVs membranes, characterized by their lipid bilayer structure, can spontaneously fuse with various lipid membranes. Yang et al. have introduced an innovative biocompatible technique known as membrane-editing, which utilizes virus-mimetic fusogenic exosomes to directly transfer functional membrane molecules into cell membranes. This method allows small EVs to target specific tissues effectively. This delivery method based on fusogenic- small EVs also enables small EVs to target specific tissues [36].

Nanoformulation techniques

Utilizing nanoformulation techniques enhance the loading capacity of MSCs-small EVs for therapeutic agents. Research has shown that BMMSC-small EVs are selectively taken up and accumulated by neuronal cells in pathologic sites, whereas the control groups exhibited only diffusive migration with subsequent clearances within 24 h. Compared to computed magnetic resonance imaging (MRI), tomography (CT) imaging provides higher tissues resolution and avoids radiation exposure in the neuronal system [6].

Jia et al. demonstrated the loading of superparamagnetic iron oxide nanoparticle (SPION) into small EVs conjugated with a myelin basic protein-1 (MBP-1) targeting peptides and encapsulated curcumin. This approach achieved targeted MRI imaging and treatment for brain gliomas [37].

In another research, MSCs -small EVs were constructed by modifying their surfaces and loading them with SPIONs. Cell-penetrating peptides (CPPs) and TNF- α (CTNF α) were anchored to the small EVs containing SPIONs. These engineered small EVs exhibited targeted antitumor effects and significantly suppressed cancer cells growth by inducing apoptosis through the TNF receptor I signaling pathways in both invitro and in-vivo melanoma subcutaneous tumor models [12].

Furthermore, Pascucci et al. utilized mice MSCs-small EVs to load paclitaxel (PTX) through cells incubation with PTX, finding that these MSCs produced MSCs-Exo

containing a high amount of PTX, which suppressed pancreatic tumor cells growth [38].

Size and shape optimization

The size and shape of MSC-small EVs significantly influence their bio-distribution and uptake by target cells. Engineering MSCs to produce small EVs of specific sizes can optimize their therapeutic properties. Studies indicate that smaller EVs may enhance tissue penetration, while larger exosomes might be more effective at delivering cargo to specific sites. Adjusting culture conditions or utilizing microfluidic devices can help achieve the desired size and shape of these small EVs [33].

Thermal and chemical stabilization

Physical modification can also enhance the stability of MSC-small EVs under physiological conditions. Techniques such as addition of stabilizing agents or lyophilization can protect MSC-small EVs from degradation during storage and transport. Ensuring the stability of MSC-small EVs is critical for maintaining their therapeutic efficacy over time [33].

Targeted release mechanisms

Techniques such as freeze-thaw cycles, and sonication electroporation can be employed to facilitate the incorporation of proteins, nucleic acids, or drugs into small EVs. These Methods can enhance the efficiency of cargo loading, ensuring that small EVs deliver a higher concentration of drug agents [33].

Furthermore, incorporating responsive materials into the surface of small EVs enable controlled drug releases in response to specific stimulation, such as enzymatic activity, temperature fluctuations, and pH changes. This method allows for precise delivery of drug agents at the target region, promoting treatment outcomes while minimizing adverse effects [33].

Chemical modification

Chemical modifications can display various synthetic or natural ligand receptors on the surface of small EVs using different techniques, broadly classified into covalent and non-covalent modifications.

This strategy involves altering the chemical properties of exosomes to improve their stability, loading capacity, targeting ability, and overall therapeutic efficacy [2]. Here are some key strategies for chemical modification:

Lipid composition alteration

Modifying lipid composition of small EVs significantly enhance their stability and surface properties. The incorporation of lipid-based compounds or specific lipids may increase small EVs resilience against degradation and

improve the ability to fuse with target cells membrane [2].

Hydrophobic interactions enable direct insertion of targeting moieties into small EVs membranes. Functionalized phospholipids can be incorporated by simple incubation using this strategy [39]. This allows for loading small EVs with small lipophilic drugs, including anti-inflammatory and chemotherapeutic agents. Commercial applications include small EVs membrane stains like BODIPY TR ceramide and DiI, which require coinubation under conditions favoring hydrophobicity [40].

Furthermore, in a study by Antes et al., a novel method called “cloaking” was introduced to successfully integrate targeting molecules onto the membrane of small EVs using a surface anchoring platform.

This cloaking method consists of three components: a conjugated platform molecule of streptavidin, a PEG spacer, and a phospholipid membrane anchor made from DMPE. By leveraging this system, any biotinylated molecule can be attached to the platform to decorate the small EVs. Vandergriff et al. demonstrated that by conjugating small EVs with a homing peptide via a dioleoyl-phosphatidylethanolamine N-hydroxysuccinimide (DOPE-NHS) linker achieved enhanced therapeutic outcomes and increased homing efficacy to the injured site of heart were achieved [41].

Currently, electroporation stands as the most widely used method for loading exogenous “cargo” into extracellular vesicles. This technique operates on the principle of applying an external electric field to create small and repairable pores in the phospholipid bilayers, allowing various small molecules to enter the EVs under the influence of the electric field, facilitating drug loading. Doxorubicin as anticancer drug can be efficiently delivered into MSCs-small EVs via electroporation, significantly reducing cancer growth rates in mice models of colon and breast cancer [11].

In a study by Liang et al., norcantharidin (NCTD) as exogenous anticancer drug was successfully loaded into MSCs-small EVs via electroporation method. These engineered MSCs-small EVs were utilized for treating mouse models of liver cancer, taking advantage of the natural liver-targeting properties of the exosome [42].

Chemical conjugation of therapeutics

Small EVs can be chemically modified to covalently attach therapeutic agents, such as small molecules, peptides, or nucleic acids. This chemical conjugation allows for controlled release of the drug at the target site, utilizing techniques like click chemistry or carbodiimide coupling to facilitate the attachment of therapeutic agents to small EVs membranes or their cargo [43]. Furthermore, Compared to biological modification approaches,

chemical reactions offer greater efficiency, simplicity, and speed [44].

Jia et al. employed click chemistry to attach the glioma-targeting peptide (NRP-1) as a specific ligand of RGE-peptide to the surface of MSC-small EVs. The modified small EVs successfully crossed the blood-brain barrier, aiding in the precise identification of gliomas [37].

Pascucci et al. utilized MSCs-small EVs to load paclitaxel (PTX) by incubating MSCs with PTX. They found that the MSCs produced small EVs containing a high concentration of PTX, which effectively inhibited the growth of pancreatic tumor cells [38]. Some materials utilized for chemical conjugation of exosomes include:

Polymer coating

Coating small EVs with biocompatible polymers can enhance their stability and circulation time in the bloodstream. Key strategies for Polymer Coating of physical modification include:

Polyethylene glycol (PEG) PEG creates a hydrophilic shield around the small EVs, minimizing protein adsorption and inhibiting immune recognition. This prolongs their half-life and increases biodistribution [2].

For instance, PEGylated liposome was incorporated into the surface of small EVs through hydrophobic interactions, significantly prolonging their circulation time and improving their targeting capability to mouse neuroblastoma cells [45].

Poly (lactic-co-glycolic acid) (PLGA) Li et al. (2018) found that bone regeneration was improved by MSC-small EVs due to their osteo-inductive properties and capacity to enhance cells homing and migration when used alongside PLGA scaffolds, which are well-known as biodegradable and biocompatible materials [46].

In addition, Xu et al. developed an innovative-scaffold by electrospinning copper-based MOF with PLGA scaffolds and immobilizing BMSCs-small EVs onto fibrous polymer meshes. This small EVs -laden scaffold, featuring a unique nanostructure, demonstrated a dual cooperative release of copper ions and small EVs, promoting angiogenesis and osteogenesis. This approach provided therapeutic effects and facilitated cells-free bone regeneration [47].

Chitosan (CS) In a study by song et al. (2024), a multi-functional hydrogel was developed using gallic acid-conjugated CS and oxidized hyaluronic acid for the sustained releases of hypoxic BMMSC-small EVs. The durability and stability of the small EVs, combined with the slow degradation of the hydrogel, facilitated their gradual release into the wound environment [48].

A study investigated the use of tissue inhibitor of metalloproteinase 2 (TIMP2)-enriched BMSC- small EV encapsulated in CS hydrogels to inhibit angiogenesis in cholangiocarcinoma. By targeting both Wnt/ β -catenin signaling pathway and angiogenesis, this multi-faceted approach offered a more effective and precise therapeutic intervention [49].

Wu et al. (2024) created a wound dressing comprising chitosan hydrogel and AdMSC-small EVs using a physical mixing method. The study revealed that the degradation rates of the chitosan hydrogel was consistently lower than the release rates of the small EVs. This indicates that the release of the small EVs is not solely dependent on the hydrogel's degradation but is also influenced by the diffusion of the small EVs from the hydrogel matrix [50].

Polycaprolactone (PCL) Rezaie et al. demonstrated that MSC-microvesicles can be encapsulated in polycaprolactone (PCL) nanofibers for extended release. This sustained release maintains an apoptotic effect on cancer cells cultured on PCL- microvesicles nanofibers, indicating that MSC- microvesicles can serve as in situ inhibitors instead of traditional chemotherapeutics. Additionally, PCL nanofibers enhance the anticancer properties of MSC-small EVs through prolonged release [51].

Similarly, Lashkari et al. (2023) developed PCL and gelatin nanofibers that were electrospun directly onto a collagen-alginate hydrogel seeded with MSC. Their result indicated that re-epithelialization, neovascularization, and collagen fibers -remodeling were significantly promoted in the cells-seeded bilayer scaffold and nanofibers compared to the group of control [52].

Alginate Chahsetareh et al. (2024) reported that an alginate hydrogel-PCL/gelatin nanofiber composite scaffolds containing MSCs-small EVs facilitated sustained release for the regeneration of tympanic membranes perforations [53].

Wu et al. bioprinted a bioscaffold using gelatin and sodium alginate loaded with MSC and nitric oxide to enhance burn wound healing. The MSCs were evenly distributed within the hydrogels, and SEM images showed that the scaffold's grid-like structure supported MSC growth. The scaffold also demonstrated a sustained release of nitric oxide, ultimately improving wound healing by promoting epithelialization, collagen deposition, and neovascularization [54].

Gelatin Lu et al. developed a composite hydrogel using hyaluronic acid methacrylate (HAMA) and gelatin methacrylate (GelMA), which was coated with growth factors and incorporated nanohydroxyapatite (nHAP) to enhance its mechanical properties. They encapsulated Urine stem cells (SCs) -small EVs and systematically characterized

them to evaluate their angiogenic and osteogenic capabilities in-vitro. The GelMA-HAMA/nHAP composite hydrogels demonstrated excellent controlled release performances and suitable mechanical properties [55].

On the other hand, Hao et al. created a GelMA matrix to encapsulate SCs-small EVs to maximize their utilization in-vivo. The long-term and sustained release of SCs-small EVs significantly enhanced bone regeneration by osteogenesis, vascularization, immune regulation, and promoting innervation [56].

Poly(lactic acid) (PLA) Li et al. (2018) found that bone regeneration was significantly enhanced by MSC-small EVs due to their osteo-inductive effects and ability to promote cells migration and homing. When combined with PLA-PLGA scaffolds, which are widely recognized as biocompatible and biodegradable materials, the MSC-small EVs contributed to improved regenerative outcomes [46].

These studies underscore the significance of polymers in engineering MSC-small EVs and enhancing their therapeutic potential across various biomedical applications.

Targeting ligands attachment

Chemical modification enables the attachment of targeting ligands, such as aptamers, peptides, and antibodies to the surface of small EVs. These ligands can specifically bind to target cells' receptors, enhancing the efficiency and selectivity of drug delivery, particularly for tumor cells or inflamed tissue [2].

The LJM-3064 aptamers, known for its strong affinity for myelin, has been shown to induce myelin regeneration. Hosseini developed a LJM-3064 aptamer that covalently bonds to the surface of small EVs using EDC/NHS chemistry. This aptamer-small EVs biological conjugate serves as an effective preventive measure to reduce neurodegeneration and inflammation in multiple sclerosis disease [57].

pH-responsive modifications

Incorporating pH-sensitive chemical moieties into small EVs enables controlled drug release in response to the acidic microenvironment of tumors or inflamed tissues. This strategy enhances the therapeutic efficacy of the cargo by ensuring that drugs are released specifically at the target site, minimizing systemic side effects [2].

Studies have illustrated that small EVs modified with pH-sensitive moieties can effectively deliver chemotherapeutic agents, such as doxorubicin, directly to tumor cells, significantly improving the treatment outcomes while reducing adverse effects [58].

A sophisticated hydrogel system designed for the dual delivery of small EVs and antimicrobial effects was developed by Wang et al. (2019). This hydrogel, composed of poly- ϵ -L-lysine, hyaluronic acid, and pluronic F127 is

pH-sensitive, where the MSCs-small EVs release rate was faster in acidic pH conditions than in neutral pH. The regeneration of skin over the wound site, treated with MSC-small EVs -loaded hydrogel, was more quickly than at the site treated with free small EVs over 21 days [59].

Crosslinking for stability

Chemical crosslinking can stabilize MSC-small EVs and their cargo. Using crosslinking agents creates a more robust structure that protects the contents of exosomes from degradation, which is crucial for maintaining the therapeutic efficacy of the delivered agents over time [2].

Li et al. demonstrated that a genipin-crosslinked hydrogel infused with MSC-small EVs promotes full-thickness skin wounds healing in a mouse model. This research showed that the MSC-small EVs -loaded hydrogel significantly enhanced wound closure, rates of re-epithelialization, and collagen deposition at the wound regions. Additionally, a higher number of skin appendages were found in wound treated with the MSC-small EVs -loaded hydrogel, highlighting its potential for skin regeneration [60]. Figure 1 summarizes the strategies for Engineering MSC-small EVs as drug delivery for cancer treatment.

The labeling and imaging of engineered small EVs

To effectively observe the dynamics of engineered small EVs in vivo, suitable imaging strategies are essential. These strategies encompass various labeling techniques, reporter systems, and microscopy methods [61, 62].

Labeling methods, including luminal dyes, lipid dyes, and genetic labeling, are crucial for visualizing engineered small EVs. Commonly used lipid dyes, such as MemGlow, PKH67, and DiR/DiD, have long half-lives that can mask small EVs degradation after cellular uptake, making them less suitable for long-term cancer studies [63]. In contrast, genetically encoded reporters—both fluorescence and bioluminescence—can be successfully integrated into engineered small EVs. Proteins like the bioluminescent ThermoLuc, RFPs, and GFPs are often loaded into small EVs, with CD63 being the most widely used marker. Bioluminescent reporters, such as gLuc-lactadherin, provide a high signal-to-noise ratio and are recommended for whole-animal imaging in small models [64, 65].

The administration method is also significant. Most in-vivo experiments rely on intravenous injections, while others may use intra-tumoral, intradermal, or intraperitoneal injections [66, 67].

Live imaging techniques are crucial for studying engineered small EVs in-vivo. Various microscopy methods, including confocal structured illumination microscopy (SIM), laser scanning microscopy (CLSM), and stochastic optical reconstruction microscopy (STORM), can

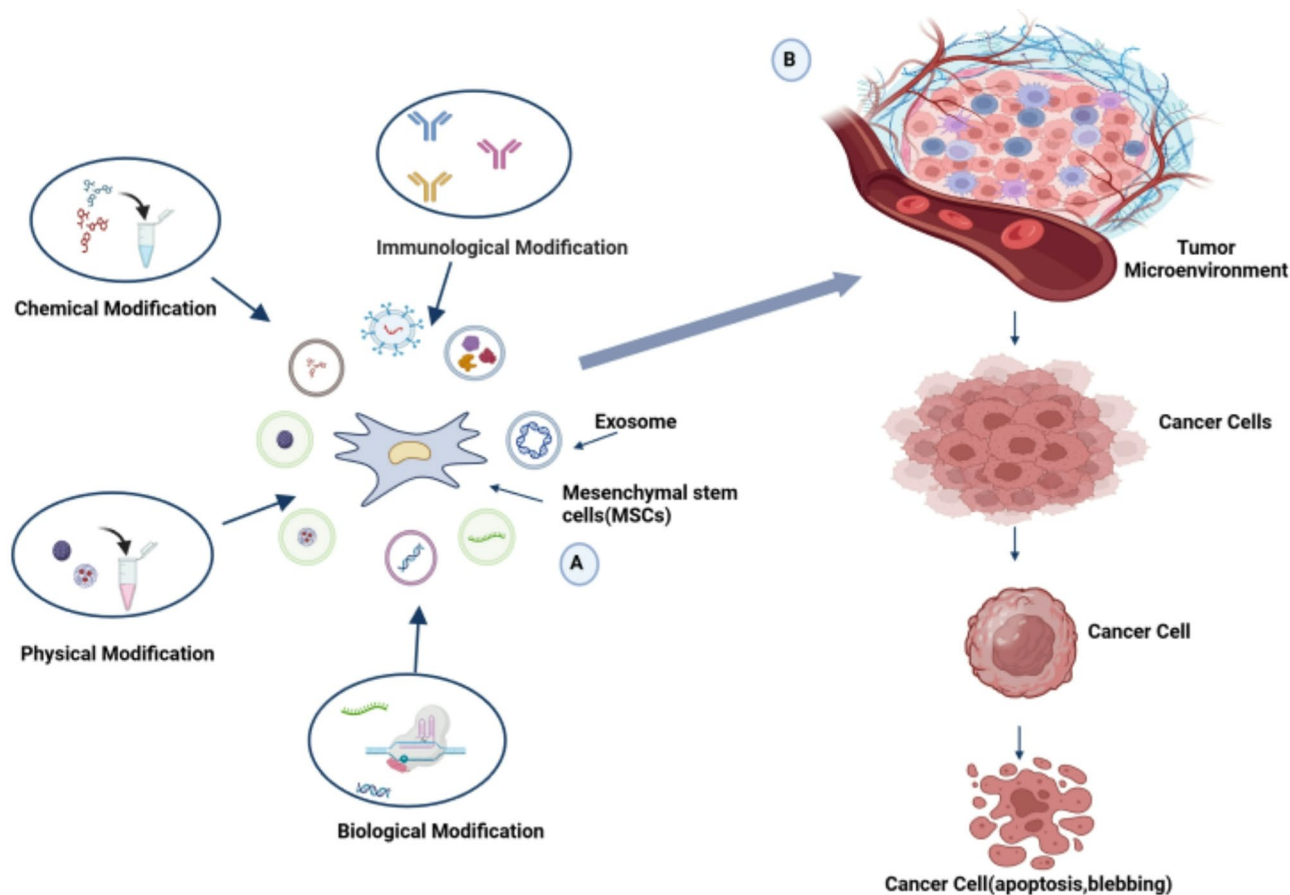


Fig. 1 Provides an overview of the various strategies employed for engineering MSC- small EVs as (A) drug delivery systems for (B) cancer treatment. Created with BioRender.com

effectively track small EVs uptake and distribution in real time. Although MRI may also be utilized for detecting biological activities, combining bioluminescent proteins with CLSM or SIM shows great promise for live imaging of engineered small EVs [68, 69].

In summary, labeling and imaging engineered small EVs are crucial for advancing our understanding of small EVs biology and improving their applications in drug delivery. There is no universal reporter or microscopy method; the choice should depend on the specific biological questions and the imaging technologies available. By employing various techniques, researchers can gain insights into the behavior, targeting, and therapeutic potential of exosome-based systems, paving the way for innovative therapeutic strategies [61, 62].

Strategies for engineering MSC- small EVs as improved therapeutic tools

The transformation of mesenchymal stem cell-derived exosomes into better therapeutic tools involves several strategies to improve their efficacy in various treatments. Herein, the potential approaches to integrate MSC- small EVs into different therapeutic modalities are presented.

Chemotherapy

The approaches may be incorporated into the development of MSC- small EVs to further enhance their therapeutic efficacy in oncological therapies. These could be engineered in such a way as to act as targeted delivery systems for chemotherapeutic agents, increasing their effectiveness and reducing adverse effects through precise drug delivery to cancerous cells [70]. In addition, MSC- small EVs have been found to influence drug resistance mechanisms in cancer cells, as exemplified by the increase in resistance to chemotherapy in gastric cancer cells through specific signaling pathways. This knowledge can be used to design exosomes targeting the reduction in resistance and improvement of therapeutic outcomes [16]. Furthermore, some techniques in biological engineering can be harnessed for encapsulating chemotherapeutic agents and expressing specific targeting ligands to increase the efficiency of small EVs in tumor targeting and pharmaceutical compound delivery [71].

Recent advances in engineering strategies of MSC- small EVs for cancer treatment manifest a number of innovative approaches. One of the main developments in this field was represented by the targeted delivery of

microRNAs. Such microRNAs as miR-145 were delivered successfully by MSC-small EVs into the cells of one of the most frequent types of cancers—breast cancer. Moreover, this approach can offer a completely new therapeutic direction, showing effectiveness in metastasis inhibition and inducing apoptosis of these cells [72].

Moreover, researchers are focusing on bioengineering in order to enhance the effectiveness of MSC-small EVs in inducing target improvements and therapeutic effects. These could be modifications such as loading of small EVs with antitumor drugs or non-coding RNAs which enhance the stability and potency of exosomes in tumor treatments [71]. Furthermore, nanobiotechnology development enabled an enhancement of MSC-small EVs for improved drug delivery. Such modified exosomes will be able to be loaded with therapeutic agents for direct delivery inside tumor cells, using natural features for targeting [73].

Radiotherapy

Radiation therapy is a core method under the therapeutic arsenal for cancers and is heavily applied for various forms of cancers like prostate, lung, and head and neck cancers [74]. An estimate suggests around 50% of all cancers are treated using the treatment by radiation, contributing about 40% towards therapeutic treatment for the cure of the disease. However, many cancers possess radioresistance owing to complex biological processes, thus lessening the treatment efficiency and possibly causing metastasis and cancer recurrence [75]. Contemporary studies also showed the ability of irradiation to alter the exosomes' composition and the mode of their secretion. Irradiation-induced exosomes cause radioresistance and induce the phenomenon of the radiation-induced bystander effect [75]. For instance, hypoxic oral squamous cell carcinoma (OSCC) cell-derived exosomal miR-340-5p has the ability to promote radioresistance by directly preventing radiotherapy-induced apoptosis and facilitating DNA repair through KLF10 [9]. In addition, exosomes derived from MSCs were also demonstrated to induce the development of cancer radioresistance. In cell lines and tissues from nasopharyngeal carcinoma, microRNA miR-34c downregulation decreases radioresistance by inducing radiotherapy-caused apoptosis through the modulation of the β -catenin under both the in vitro and the in vivo settings. In addition, MSC-derived exosomes containing high levels of miR-34c strongly increase the levels of radiotherapy-caused apoptosis in the context of nasopharyngeal carcinoma [9]. In addition, MSC-small EVs can also improve the effectiveness of radiotherapy for the inhibition of tumor growth and metastasis, as evidenced by using the melanoma mouse model. This suggests that the combination of radiotherapy and MSC-small EVs could be a potential therapeutic

intervention for the treatment of cancer patient populations [76].

Photodynamic therapy (PDT)

The combination of MSC-derived exosomes and PDT might provide a new promising approach for cancer treatment, in which therapeutic efficacy could be considerably improved. MSC-small EVs are nanosized EVs that mediate intercellular communication in nature and can be engineered for delivering therapeutic agents, including miRNAs or photosensitizers, directly to the tumor cells. These exosomes possess excellent biocompatibility and low immunogenicity; moreover, they have the capacity to modulate the growth of a tumor through delivery of regulatory molecules that could suppress or stimulate the progression of the tumor. For example, miR-145-expressing exosomes secreted by MSCs have been known to induce apoptosis in prostate cancer cells by initiating caspase pathways; others also restrict angiogenesis through the VEGF expression by targeting mTOR/HIF-1 α signaling pathway [77, 78]. MSC-small EVs in PDT, selectively inducing apoptosis of cancer cells through light-sensitive reactive oxygen species with photosensitizer, small EVs can enhance the concentration and effectiveness of the therapy. Not only will it enhance delivery of the drug, but it will also maximize anti-tumor immune activation elicited through PDT. Nonetheless, concerns regarding population heterogeneity of exosomes and their dual role in suppression and stimulation in tumors must first be resolved in order to maximize their potential in a clinical setting [14, 79, 80].

Photo thermal therapy

MSC exosomes have become the focus of a burgeoning innovative tool in using photothermal therapy for the treatment of cancer [81]. The engineering of small EVs from MSCs allows embedding photothermal agents such as ICG, which converts light into heat when exposed to a laser to induce apoptosis in cancer cells while reducing damage to the surrounding healthy cells [82]. Besides that, therapeutic agents can also be encapsulated within such engineered exosomes, enabling controlled releases induced by even light, further enhancing the general efficacy of any treatment [83]. Treatment techniques such as ultrasound and electrical stimulation have, to date, enhanced yield and functionality in MSC-small EVs while the development of nanotechnology greatly facilitated the isolation and functional modifications of the same for application in clinical practice [84]. The engineered small EVs incorporated in PTT not only introduce a targeted therapeutic modality for cancer but also align with the tenets of precision medicine, with a diminution in traditional treatment-related side effects being an added benefit. There is a necessity for additional studies to normalize

the processes for engineering and analysis of the biological interactions in the tumor microenvironment to maximize such novel therapies for clinical applications.

Sonodynamic therapy (SDT)

The combination of sonodynamic therapy (SDT) with engineered MSC-small EVs is a hopeful method for improvement in therapeutic efficacy for cancer. Engineered exosomes have been proven to be effective carriers for ICG, a sonosensitizer, for site-specific delivery to neoplastic cells with reduced systemic toxicity [85]. By engineering small EVs with ligands that target tumors, scientists have increased their density in cancer locations and therefore taken them up more effectively with sonosensitizers and maximized the efficacy of sonodynamic therapy [86]. Research shows that chemotherapy drug small EVs paired with sonosensitizer paclitaxel exhibit the ability to reverse drug resistance. The ultrasound treatment generates ROS more effectively, leading to advanced cancer cell damage. Sonosensitization presents extensive therapeutic potential for curing complex cancer types, including glioblastoma. Small EVs agents have been modified to penetrate blood-brain barriers, allowing their therapeutic impact to occur [87]. The clinical application remains limited due to unsorted scalability issues and the absence of standardized protocols and regulatory approvals. The combination of small EVs sourced from mesenchymal stem cells and sonodynamic therapy stands as a leading potential technology for cancer treatment despite remaining largely untapped.

Genetic modifications

Such features of MSCs and, consequently, their derived exosomes, which are called MSC-small EVs, have made them an attractive tool in cancer therapy: low immunogenicity, high stability, and tumor-site targeting. More recently, with developments in genetic modification techniques, this therapeutic efficacy has been further enhanced, making MSC-small EVs an efficient vehicle in cancer treatment. The overexpression of therapeutic genes is one of the ways adopted to improve the therapeutic functionalities of MSC-small EVs. Genetic modification of MSCs with genes encoding proteins against cancer, like tumor suppressor genes or immunomodulatory factors, has resulted in an improved anti-tumor response. MSCs engineered to express TRAIL (TNF-related apoptosis-inducing ligand) showed the induction of apoptosis in cancer cells and reduced tumor growth in different models [71, 88]. Another powerful approach is the knockdown of oncogenes in MSCs. The silencing of such oncogenes might prevent tumor growth and metastasis that may be related to the dual role that unmodified MSCs can play in promoting tumor progression. Scientific studies demonstrate that genetic

modifications which decrease oncogene expression lead to lowered metastasis risk and better therapeutic results [79, 89]. MSC-small EVs therapy development also raises crucial questions about targeted delivery systems. By engineering MSCs to express particular targeting ligands for their surface researchers can enhance exosome delivery to tumor regions. Making such targeting, higher accumulation of therapeutic agents at the tumor site is obtained and simultaneously reduces off-target effects. Engineering MSCs to express chemokine receptors, such as CXCR4, it was noted that MSCs exhibited enhanced migration toward tumors [73, 90]. Applications of genetically modified MSC-small EVs are multifarious. The exosomes function as transport carriers for multiple therapeutic molecules while containing chemotherapeutic medications and microRNAs. These vesicles naturally penetrate tissues while evading immune recognition to increase their potential as drug delivery vehicles. The cancer cell growth inhibition was enhanced when paclitaxel (PTX) was loaded into small EVs as opposed to using the drug in its free form [91, 92]. Small EVs retain most characteristics of the parental MSCs responsible for their immunomodulatory impacts driving positive changes in the tumor microenvironment. Cell-free therapeutic usage of MSC-small EVs also excludes disadvantages inherent in cell-based therapies which include lineage differentiation and immune rejection [71, 88]. Despite these developments, however, there are still challenges to MSC-small EVs therapy. Unmodified MSC-small EVs sometimes foster the growth of tumors since they carry pro-tumorigenic factors; hence, these exosomes must be carefully engineered so as not to inadvertently make the tumor more progressive [79, 89]. Protocols for standardized isolation and application of MSC-small EVs in clinical settings must also be designed. Thus, genetic modification of MSC-small EVs holds great promise in enhancing cancer therapy efficacy via targeted drug delivery with reduced side effects and superior therapeutic outcomes. It is expected that more research shall be required to overcome the challenges presently faced, and they may then be completely realized in clinical applications [73, 90].

Recent advances in engineering MSC-sEVs

MSC-small EVs are considered therapeutic agents due to their ability to mediate intercellular communication through the transfer of proteins, mRNAs, miRNAs and DNAs, as well as their favorable low immunogenicity and tumorigenicity. Their targeting ability and therapeutic utility have subsequently been improved by new technical approaches [71, 93, 94].

Precise genetic editing using CRISPR/Cas9

CRISPR/Cas9 technology is a new genome editing platform based on bacterial adaptive immunity that allows efficient and specific editing of genomic sequences. Delivery of Cas9 nuclease and guide RNA into MSCs permits specific editing of genes that are involved in the biogenesis of exosomes, cargo loading, or ligand targeting, thus engineering MSC-small EVs for improved therapeutic functions or targeting of a disease [95]. The technology represents a very advanced method of engineering MSC-small EVs with high precision, low cost, as well as broad applicability to various organisms. Recent advancements confirm that CRISPR/dCas9 activation accurately regulates extracellular vesicle cargo by overexpressing therapeutic genes independent of full-length transgene delivery and offers better and reproducible exosome preparations. For instance, CRISPR activation of the TSG-6 gene enhanced MSC-EVs protein level significantly without impairing EV morphology and MSC differentiation capacity. It regulates EV cargo at the point of origin and offers a more regulated and reproducible way of enhancing therapeutic potency as opposed to the utilization of traditional gene overexpression [96–99]. Furthermore, CRISPR/Cas9 editing of MSCs provides the advantage of compositional regulation of secreted small EVs towards improved target ability and therapeutic application [100, 101]. For example, CRISPR/Cas9 knockout of specific genes of MSCs was shown to regulate the composition of the extracellular matrix and improve regenerative ability. Furthermore, small EVs themselves can be utilized as native nanocarriers for delivery of CRISPR/Cas9 tools towards the target cells free of the constraints of delivery efficacy and immunogenicity of viral or synthetic vectors [101]. Cumulatively, these advances position CRISPR/Cas9 as a multifunctional, scalable platform technology for precise MSC exosome engineering having far-reaching applications in precision medicine, as well as cancer therapy. Small EVs delivery of CRISPR/Cas9 also increases specificity of targeting, as well as decreases off-targeting, and therefore represents a promising approach to therapeutic use in the future [102, 103].

Surface modification via novel chemical methods (click chemistry)

Click chemistry is a bioorthogonal and versatile approach to chemical modification of the surface of MSC-small EVs without compromising their integrity and bioactivity. Efficient and selective chemistry can be employed to conjugate targeting entities or functional groups with exosome membranes, promoting their stability, biodistribution, and cell-specific internalization. Click chemistry may be employed for the conjugation of ligand, imaging molecule, or therapeutic molecule to the MSC-small EVs,

making the MSC-small EVs-based therapies more targeted [104, 105].

Hybrid technologies: combining nanoparticles with MSC-small EVs

Integration of nanotechnology with MSC-small EVs makes it easier to prepare multifunctional therapeutic platforms. Hybrid platforms of nanoparticles (gold, magnetic, or polymeric nanoparticles) and MSC-small EVs may provide increased drug loading, controlled drug release, and targeting selectivity [106, 107]. Hybrid nanostructures may benefit from the natural biocompatibility and targeting ability of exosomes as well as the physicochemical behavior of nanoparticles, which can open new frontiers for next-generation theranostic applications for targeted drug delivery as well as tissue regeneration.

These new engineering methods collectively position MSC-small EVs as multifaceted, efficient, and individualized Nano therapeutics with extensive applications for tissue regeneration, immunomodulation, and therapeutic uses for complicated diseases [108, 109].

Preclinical and clinical studies demonstrating the efficacy of engineered small EVs

MSC-small EVs as a cell-free therapeutic material represents an innovative and promising approach to cancer treatment, offering targeted delivery of therapeutic molecules and modulating the tumor microenvironment to enhance antitumor efficacy. Their potential for commercial scalability and their ability to be functionalized for targeted tumor therapy enhance their appeal [79]. To establish small EVs as effective drug carriers in clinical settings, it is essential to develop biocompatible and economically viable small EVs that are effective and well-tolerated in vivo. This necessity has prompted researchers to investigate exosome and their role in physiological and pathological conditions more fully. Progress in the clinical use of Exo has been relatively slow. However, Phase I clinical trials involving various cancers have demonstrated the safety of exosome-based vaccines, suggesting their potential as a new approach to cancer immunotherapy [110]. The first clinical of MSCsmall EVs was conducted in 2014. Most clinical trials using MSCsmall EVs focused on chronic diseases, immunity disorders, and acute ischemic stroke [99]. Notably, only one clinical trial has investigated MSC-small EVs in cancer treatment. M.D. Anderson Cancer Center initiated a clinical trial (NCT03608631) using MSC-small EVs containing K-RAS G12D siRNA(iExos) to treat metastatic pancreatic cancer. The clinical-grade GMP iExos underwent extensive in vitro and in vivo testing, demonstrating their ability to suppress oncogenic Kras and significantly enhance the survival of various mouse models with pancreatic cancer [111]. They also developed a scalable method for

producing these exosomes using a bioreactor. The primary objectives of the trial are to identify the maximum tolerated dose (MTD) and dose-limiting toxicities (DLT) of iExosomes (iExos) in patients with metastatic pancreatic ductal adenocarcinoma (PDAC) harboring the K-RAS G12D mutation. Secondary objectives include evaluating the pharmacokinetic profile, overall response rate, disease control rate, progression-free survival (PFS), and overall survival (OS) of iExos. The study also explores the detection of DNA and RNA showing K-RAS G12D sequence in tissue and blood samples (www.clinicaltrials.gov).

More recently in-vitro studies by Jafarpour et al. revealed that MSC-small EVs loaded with PTX and cisplatin (CIS) facilitated efficient cellular uptake induced higher toxicity and apoptosis in the 3D multicellular spheroids model of breast cancer cell lines [112]. Yu et al. used hUC-MSC-small EVs as delivery vehicles for an anti-miR-146b-5p antisense oligonucleotide (PMO-146b) ePPMO-146b was effectively taken up by SW620 colorectal cancer cells and significantly inhibited their proliferation and migration. The PMO-146b was enriched in tumor tissue of a xenograft mouse model of colorectal cancer, inhibiting epithelial-mesenchymal transition [113].

The study by Liang et al. demonstrated that the encapsulation of norcantharidin (NCTD) in BMMSC-small EVs promoted the uptake of NCTD by HepG2 cells, reduced their proliferation, and increased apoptosis. In animal models, BMMSC-small EVs demonstrated significant anti-tumor efficacy and repaired damaged liver tissues in liver Sect [42]. In a study conducted by Liu et al. BMSC-EV-derived let-7i showed great potential as a therapeutic approach for lung cancer by targeting the KDM3A/DCLK1/FXYD3 axis [114]. Table 1 presents in vitro and preclinical experiments, while Table 2 shows the clinical trials of engineered MSCs-small EVs in cancer disorders.

Challenges of MSC-small EVs therapy and future perspectives

Exosomes secreted by MSCs have been demonstrated to be responsible for various types of therapy resistance in different cancers. Small EVs are made up of a broad range of bioactive molecules that can serve as drug response noninvasive biomarkers to measure the effectiveness of therapy and drug resistance in individuals. With continuous examination of bioactive compounds, medical experts can gain a more specific measurement of treatment response to allow the choice of the best treatment approaches specific to a patient. Additionally, the acknowledgment of MSC- small EVs as a cell-free therapy is revolutionizing the approach to treating cancer. Engineered MSC-small EVs have the potential to deliver

synthetic biocomposites that have strong anticancer activity as well as anticancer agents. To date, clinical trials have started or are underway to examine the potential of MSC-small EVs to serve as a therapy modality.

Recent evidence highlights the crucial role of cancer stem cell (CSC)-derived small EVs in conferring resistance to chemotherapy and immunotherapy, as well as maintaining the stem-like characteristics of tumors. CSC-small EVs convey a wide array of bioactive cargoes, including proteins, microRNAs, and checkpoint factors such as PD-L1, which create a pro-immunosuppressive environment within the cancer. For instance, small EVs - PD-L1 produced by the CSCs induces exhaustion of CD8+ cytotoxic T cells [115] and allows cancer cell lines to evade killing by the immune system, thereby hindering the efficacy of immunotherapies targeting the PD-1/PD-L1 axis [116]. Furthermore, small EVs released by the CSCs possess the capacity to carry microRNAs and signal transduction factors that activate pathways of drug-resistance, self-renewal, and epithelial-mesenchymal transition (EMT), thereby maintaining the stem-like phenotype intact and exacerbating the malignancy of cancer [117]. The two-way role of CSC-small EVs is thus demonstrated by these features, where they contribute to therapeutic resistance as well as the maintenance of the cancer-initiating population of cells, and thus hold a crucial role in fortifying the effectiveness of cancer therapy [118].

Despite the knowledge of MSC-small EVs during the last decades, several important questions and limitations still exist for clinical application. One big issue is the lack of standard classification for MSC-small EVs. Though recently it has been suggested to propose protocols for the isolation and characterization of six different subpopulations of extracellular vesicles from tissues, a uniform international classification system for MSC-derived extracellular vesicles has not been determined. This inconsistency leads to the concern that previous findings have been affected by different methods of isolation and purification [119]. The development of strong criteria and protocols for the MSC-small EVs' purity, isolation, and storage is imperative to allow their practical applications within the clinics. Various methodologies of MSC isolation and purification produce significantly variable subpopulations of MSC-small EVs with distinct contents, properties, and bioactivities [120]. Continuing the basic research using MSC-small EVs, new research on their extraction, purification, and storage methods according to GMPs should be done and standardized. Large-scale production of MSC-small EVs is another major problem in the application of these vesicles in a clinical setting. Until now, several strategies effectively were developed for the isolation of -small EVs, such as PEG isolation, DEP separation, and DLD separation.

Table 1 The in-vitro and preclinical experiments of engineered MSCs-small EVs in cancer diseases

Tumor models	Putative name	Clinical therapy	Mechanisms	Ref.
Breast cancer	miR-379 -small EVs	GT	Elevate miR-379. decrease COX-2 mRNA and proteins in-vitro and in-vivo.	[135]
Hepatocellular cancer	miR-125b-small EVs	GT	Overexpress miR-125b in HCC cell. Induce cells cycle arrest. Prevent growth, migration/invasion.	[136]
Gastric cancer / Colorectal cancer / lung cancer	CXCR4 ^{high} small EVs / si-Survivin	GT	Bind to the highly expressed SDF-1 on the cancer surface. Decrease the Survivin genes in cancer cells in vivo, leads to a significant suppression of tumor growth.	[137]
Breast cancer	miR-222/223-Eexos/ Carboplatin	GT	Carboplatin-based therapy and increase host survival	[138]
Glioma	miR-29a-3p-small EVs	GT	Prevent migration and VM formation. Target ROBO1 through miR-29a-3p	[139]
Breast cancer	miR-23b /MARCKS/ DOX	GT	Improve cancer cells dormancy in a metastatic niche	[15]
Breast cancer	miR-21-5p /S100A6/ DOX	GT	Promotes chemo-resistance	[140]
Acute myeloid leukemia(AML)	OCT4, NANOG and SOX2/ Ara-C	GT	Promote AML cells proliferation, invasion and chemo-resistance through upregulation of S100A4.	[141]
Breast cancer	miR-1236/ Wnt/ β -Catenin/ cisplatin	GT	Reduces Resistance of cancer cell to cisplatin by inhibiting the Wnt/ β -Catenin and SLC9A1 signaling.	[142]
Ovarian cancer	microRNA-146a-small EVs	GT	Increases the sensitivity of tumor cells of ovarian to docetaxel and taxane through a LAMC2-mediated PI3K/Akt axis cross-prime cancer-reactive CD8 + T-cells response.	[142]
Hepatocellular carcinoma	miR-451a-small EVs	GT	MiRNA-mediated EMT and cancer stem cell in cancer chemoresistance	[143] [79]
Breast cancer	miR-21 and miR-34a	GT	Supported breast cancer cell proliferation and metastasis possibly by transferring miR-21 and miR-34a	[144]
Breast cancer	MSC544 -small EVs -Taxol-	CT	Taxol-loaded MSC544 sEVs showed superior cancer-reducing abilities	[79]
Pancreatic cancer cells	honokiolo into MSC-EVs	CT	Increase cellular uptake of Honokiolo and its antitumor efficacy	[145]
Pancreatic cancer	BM-MSCs-small EVs-PTX- gemcitabine monophosphate (GEMP)	CT	sEVs delivery platform showed superiorities in homing and penetrating abilities	[146]
Thyroid cancer	New TKla-small EVs	RT	Acting as nano-transporters, these carriers efficiently deliver tyrosine kinase inhibitors (TKIs) to thyroid tumor cells, restoring iodine avidity and enhancing the tumor's sensitivity to radiotherapy.	[147]
Non-small cell lung cancer (NSCLC)	Small EVs - miR-101-3p	RT	Promotes radiotherapy sensitivity in NSCLC by regulating DNA damage repair and autophagy levels through EZH2	[148]
Melanoma	$\gamma\delta$ -T- EV	PDT	Photosensitizer Chlorin e6 (Ce6) generates reactive oxygen species (ROS) in melanoma cell.	[149]
Bladder cancer	PEP-small EVs -CoS	PDT	Increase of ROS in bladder cancer cell. Reduce the damage of CoS QD to normal tissue.	[150]
Prostate Cancer	MVs- gold nanostars (GNS)	PTT	Show an effective delivery platform utilizing MSCs-MV -GNS.	[151]
Multinodular tumours in the pancreas regions	PEG-HGNs-small EVs	PTT	Able to reach multinodular region and phase of advanced metastatic, eradicating most cancer growth in multiple cancerous nodules in the pancreas regions.	[152]

HCC hepatocellular cancer, CTL cytotoxic T lymphocyte, TME tumor microenvironment, PDT photodynamic therapy, ROS reactive oxygen species, PTT photothermal therapy, RT radiotherapy, MRI magnetic resonance image, CT Chemotherapy

The amalgamation of various strategies of isolation can potentially deliver a very efficient means of attaining highly purified Exo isolation [121, 122]. However, this resultant approach is usually associated with increased

cost and increased process intricacy. Therefore, the development of real-time measurement and analytical strategies will remain imperative to the advancement of research and the medical application of MSC-small EVs.

Table 2 Summary of clinical trials of engineered MSCs-small EVs in cancer diseases (clinical trial.gov)

NO	Study Title	NCT Number	Status and Study Start	Conditions	Interventions	Outcome Measure	Trial Type and Enrollment
1	lExos in Treating Patients with Metastatic Pancreas Tumor with KrasG12D Mutation	NCT03608631	Active, not recruiting, 2021-01-27	• KRAS NP_004976.2:p.G12D, • Metastatic Pancreatic Adenocarcinoma, • Pancreatic Ductal Adenocarcinoma.	• Drug: MSCs-Exos with KRAS G12D siRNA	Maximum tolerated dose determined by Dose Limiting Toxicity (DLT). Overall survival. minimal residual disorders rate in high-risk patients. Progression-free survival.	Interventional NO:15
2	MSC-extracellular vesicles in Dystrophic Epidermolysis Bullosa	NCT04173650	Recruiting, 2024-08-13	• Dystrophic Epidermolysis Bullosa	• Drug: AGLE 102	Treatment-emergent Adverse Events. Wound healing improvement	Interventional NO:10
3	UCMSC-exosome for Chemotherapy-induced Myelosuppression in AML	NCT06245746	Not yet recruiting, 2025-03	• Acute Myeloid Leukemia(AML), • Neutropenia, • Anemia.	• Biological: UCMSC-Exo	Incidence of Adverse Events (AEs) and serious AEs. DLT. Maximum tolerated dose	Interventional NO:9

AML; Acute Myeloid Leukemia

In addition, the engineered exosomes should be loaded with antitumor agents and deliver them with improved efficacy. Artificial small EVs have more recently attracted great interest in being able to develop effective drug delivery systems. Artificial small EVs share the same characteristics and therapeutic prospects as their endogenous equivalents, with greater pharmaceutical acceptability and the possibility of large-scale production [123, 124].

MSC-derived exosome mimetics for the treatment of cancer are a very promising direction toward broad clinical applications in the future. For example, PTX complexed by the process of extrusion onto exosome-mimetics derived from human BM-MSC showed obvious antitumor effects during in vitro and in vivo investigations into breast cancer. Despite their natural ability to home in on tumors, it is important to improve targeting specificity since MSC-small EVs can be taken up by a variety of cell types in order to minimize any side effects brought about by targeting unintended cells. Innovative strategies involve the attachment of specific peptides or ligands to the small EVs membrane via surface engineering in the quest to enhance this targeting specificity. For this, LAMP2b has been developed as a broadly used small EVs membrane protein into which different targeting ligands can be incorporated for increased selectivity toward specific cells [125, 126].

Additional small EVs membrane targetable proteins whose expression can enhance targeting specificity toward cancer cells are the transmembrane protein platelet-derived growth factor receptor (PDGFR), tetraspanin superfamily members such as CD63, CD9, and CD81 with two extracellular loops—and lactadherin, its C1C2 domain [127, 128]. Although some improvements have been made, the number of completely new high-affinity

and specific biomolecules targeting tumor cells is still small. It requires at first place an appropriate strategy against the emerging adverse phenomena rooted in surface engineering methods. More serious problems relate to questions of the biosafety of clinical-grade MSC-small EVs due to the numerous types of potentially bioactive or harmful components incorporated.

Consequently, future work should address the need for developing effective engineering methods for eliminating or deactivating the unwanted and toxic compounds from small EVs derived from MSCs, thus assuring the safety of small EVs therapies [120]. Future endeavors will need to ensure the stabilization of the antitumor agents entrapped in MSC-small EVs after their entry into the human organism. In addition, exhaustive studies need to determine the best dosage, drug distribution, treatment schedule, and biosafety for the therapeutic use of small EVs. Up until now, much work has focused upon cell-level analysis; thus, the biosafety and effectiveness of applications using MSC-small EVs need to be evaluated by using follow-up studies over the long term and the models in vivo. Large-scale, multi-center, and long-term trials will need to confirm the therapeutic value of MSC-small EVs.

Critical perspectives

Limitations of current evidence

Much of what we know has been obtained by in vitro studies or animal models, which, while displaying anti-tumor and pro-tumor activities, have questionable translatability to human cancer due to the much more complex extracellular matrix, vasculature, heterogeneous populations of cells, and interactions between the immune system present in patients, difficult to replicate in vitro [129, 130]. Published reports range from the inhibition

of tumor growth, angiogenesis, suppression, to increased tumor growth, metastasis, and evasion from the immune system; these are perhaps a reflection of variations in tissue source, conditioning of MSCs, cancer type, and stage investigated, as well as in exosome content during various stress or physiological states. Technical and production-related issues also persist, as the isolation, analysis, and bulk production of exosomes remain cumbersome, costly, and, unlike living MSCs, exosomes are non-self-propagating entities that, having been delivered, cannot generate more small EVs, so have a finite therapeutic window [129].

Key unresolved questions and analytical interpretation

MSC-small EVs behave contextually, cancer-type, and stage-specifically, modulating tumor microenvironments in varying activity and content based on cancer type and stage [91, 131]. MSC-small EVs can suppress tumor growth through inhibition of angiogenesis by delivering miR-16 in a model but promote growth by augmenting pro-angiogenic signaling or immunity suppression in others [132]. Such bivalence makes them problematic therapeutics in areas of safety and foreseeability, as controls over MSC activity are modifiable in the tumor microenvironment, and cancer-educated MSCs have the potential to produce exosomes that promote immunosuppressive macrophage differentiation, enhancing malignancy [133]. Manipulating MSC-small EVs to deliver therapeutic payloads of choice or to avoid pro-tumorigenic actions promises to be promising, but to do so would require a better understanding of exosome biology, as well as tumor heterogeneity, to enable personalization to a cancer subtype or patient. Longitudinal data on safety, in particular of bio-distribution, potential promotion of metastasis, or off-targets in non-tumor tissue, are lacking [134]. Significantly, a recent report demonstrates MSC-small EVs play a critical role in myeloid-derived suppressor cell differentiation into macrophages controlling malignant growth, further testifying to MSC-small EVs' rich, multifarious, but paradoxically problematic, regulatory influence upon tumor microenvironments [133].

Discussion

MSC-small EVs are promising for modulating the tumor microenvironment (TME) and delivering targeted therapies by regulating pathways like Wnt/ β -catenin or PI3K/Akt, which are often dysregulated in cancers [81]. However, their dual role—either inhibiting tumor growth or inadvertently promoting progression through angiogenesis or immune evasion—necessitates precise engineering [153]. Techniques such as surface modification with tumor-specific ligands or loading with anti-tumor miRNAs can enhance specificity and mitigate risks [154].

From our perspective, tailoring MSC-small EVs to cancer type is critical, as their efficacy varies between hematological malignancies and solid tumors due to differing TME compositions [155].

The immunomodulatory effects of MSC-small EVs, driven by anti-inflammatory cytokines (e.g., IL-10), can shift the TME toward anti-tumor immunity [156]. Yet, variability in small EVs cargo due to MSC source heterogeneity (e.g., bone marrow vs. adipose tissue) complicates outcomes [157]. We propose standardized MSC-small EVs preparations and comparative studies to ensure consistent efficacy. Additionally, MSC-small EVs' potential as diagnostic biomarkers, carrying tumor-specific signatures, could complement their therapeutic role, advancing personalized oncology [158]. However, challenges like scalability and long-term safety remain barriers to clinical translation [159].

In conclusion, MSC-small EVs hold transformative potential in cancer treatment, but their application requires tailored engineering, rigorous testing across cancer models, and solutions for large-scale production.

Conclusion

MSC-small EVs is introduced for the treatment of cancer are a very promising direction toward broad clinical applications in the future. Engineered small EVs have more recently involved in developing effective drug delivery systems and targeted therapy.

They show the characteristics such as possible and potent therapeutic tools in consideration with major problems which are faced to such as process of large-scale production and also GMP regulation.

Acknowledgements

Not applicable.

Author contributions

1-5 wrote draft of manuscript. 1-2 and 6 edited and reviewed manuscript. 1 prepared figure and tables file.

Funding

This research did not receive any specific grant from funding agencies.

Data availability

No datasets were generated or analysed during the current study.

Declarations

Competing interests

The authors declare no competing interests.

Received: 4 March 2025 / Accepted: 2 July 2025

Published online: 21 July 2025

References

1. Huang D, Huang W, Liu M, Chen J, Xiao D, Peng Z, et al. Progress of mesenchymal stem cell-derived exosomes in targeted delivery of antitumor drugs. *Cancer Cell Int.* 2025;25(1):169.

2. Liang Y, Duan L, Lu J, Xia J. Engineering exosomes for targeted drug delivery. *Theranostics*. 2021;11(7):3183–95.
3. Ma Y, Sun L, Zhang J, Chiang CL, Pan J, Wang X, et al. Exosomal mRNAs for Angiogenic-Osteogenic coupled bone repair. *Adv Sci (Weinh)*. 2023;10(33):e2302622.
4. Huang X, Ding J, Li Y, Liu W, Ji J, Wang H, et al. Exosomes derived from PEDF modified adipose-derived mesenchymal stem cells ameliorate cerebral ischemia-reperfusion injury by regulation of autophagy and apoptosis. *Exp Cell Res*. 2018;371(1):269–77.
5. Lang FM, Hossain A, Gumin J, Momin EN, Shimizu Y, Ledbetter D, et al. Mesenchymal stem cells as natural biofactories for exosomes carrying miR-124a in the treatment of gliomas. *Neuro Oncol*. 2018;20(3):380–90.
6. Sun Y, Liu G, Zhang K, Cao Q, Liu T, Li J. Mesenchymal stem cells-derived exosomes for drug delivery. *Stem Cell Res Ther*. 2021;12(1):561.
7. Bellavia D, Raimondo S, Calabrese G, Forte S, Cristaldi M, Patinella A, et al. Interleukin 3- receptor targeted exosomes inhibit in vitro and in vivo chronic myelogenous leukemia cell growth. *Theranostics*. 2017;7(5):1333–45.
8. Vogus DR, Evans MA, Pusuluri A, Barajas A, Zhang M, Krishnan V, et al. A hyaluronic acid conjugate engineered to synergistically and sequentially deliver gemcitabine and doxorubicin to treat triple negative breast cancer. *J Control Release*. 2017;267:191–202.
9. Chen F, Xu B, Li J, Yang X, Gu J, Yao X, et al. Hypoxic tumour cell-derived Exosomal miR-340-5p promotes radioresistance of oesophageal squamous cell carcinoma via KLF10. *J Experimental Clin Cancer Res*. 2021;40:1–17.
10. Wang X, Chen Y, Zhao Z, Meng Q, Yu Y, Sun J, et al. Engineered exosomes with ischemic Myocardium-Targeting peptide for targeted therapy in myocardial infarction. *J Am Heart Assoc*. 2018;7(15):e008737.
11. Chen S, Sun F, Qian H, Xu W, Jiang J. Preconditioning and engineering strategies for improving the efficacy of mesenchymal stem Cell-Derived exosomes in Cell-Free therapy. *Stem Cells Int*. 2022;2022:1779346.
12. Zhuang M, Chen X, Du D, Shi J, Deng M, Long Q, et al. SPION decorated exosome delivery of TNF- α to cancer cell membranes through magnetism. *Nanoscale*. 2020;12(1):173–88.
13. Amarnath S, Foley JE, Farthing DE, Gress RE, Laurence A, Eckhaus MA, et al. Bone marrow-derived mesenchymal stromal cells Harness purinergic signaling to tolerize human Th1 cells in vivo. *Stem Cells*. 2015;33(4):1200–12.
14. Nam GH, Choi Y, Kim GB, Kim S, Kim SA, Kim IS. Emerging prospects of exosomes for cancer treatment: from conventional therapy to immunotherapy. *Adv Mater*. 2020;32(51):2002440.
15. Ono M, Kosaka N, Tominaga N, Yoshioka Y, Takeshita F, Takahashi RU, et al. Exosomes from bone marrow mesenchymal stem cells contain a MicroRNA that promotes dormancy in metastatic breast cancer cells. *Sci Signal*. 2014;7(332):ra63.
16. Ji R, Zhang B, Zhang X, Xue J, Yuan X, Yan Y, et al. Exosomes derived from human mesenchymal stem cells confer drug resistance in gastric cancer. *Cell Cycle*. 2015;14(15):2473–83.
17. Pan Q, Kuang X, Cai S, Wang X, Du D, Wang J, et al. miR-132-3p priming enhances the effects of mesenchymal stromal cell-derived exosomes on ameliorating brain ischemic injury. *Stem Cell Res Ther*. 2020;11(1):260.
18. Shen H, Yao X, Li H, Li X, Zhang T, Sun Q, et al. Role of exosomes derived from miR-133b modified MSCs in an experimental rat model of intracerebral hemorrhage. *J Mol Neurosci*. 2018;64(3):421–30.
19. Wei H, Green E, Ball L, Fan H, Lee J, Strange C et al. Proteomic analysis of exosomes secreted from human Alpha-1 antitrypsin overexpressing mesenchymal stromal cells. *Biology (Basel)*. 2021;11(1).
20. Pang J-L, Shao H, Xu X-G, Lin Z-W, Chen X-Y, Chen J-Y et al. Targeted drug delivery of engineered mesenchymal stem/stromal-cell-derived exosomes in cardiovascular disease: recent trends and future perspectives. *Front Bioeng Biotechnol*. 2024;12.
21. Vidal M. Exosomes and GPI-anchored proteins: judicious pairs for investigating biomarkers from body fluids. *Adv Drug Deliv Rev*. 2020;161–162:110–23.
22. Ahmadi M, Mahmoodi M, Shooran M, Nazari-Khanamiri F, Rezaie J. Harnessing normal and engineered mesenchymal stem cells derived exosomes for Cancer therapy: opportunity and challenges. *Int J Mol Sci*. 2022;23(22).
23. Zhang S, Jiang L, Hu H, Wang H, Wang X, Jiang J, et al. Pretreatment of exosomes derived from hUCMSCs with TNF- α ameliorates acute liver failure by inhibiting the activation of NLRP3 in macrophage. *Life Sci*. 2020;246:117401.
24. Liu K, Cai GL, Zhuang Z, Pei SY, Xu SN, Wang YN, et al. Interleukin-1 β -Treated mesenchymal stem cells inhibit inflammation in hippocampal astrocytes through Exosome-Activated Nrf-2 signaling. *Int J Nanomed*. 2021;16:1423–34.
25. Yang R, Huang H, Cui S, Zhou Y, Zhang T, Zhou Y. IFN- γ promoted exosomes from mesenchymal stem cells to attenuate colitis via miR-125a and miR-125b. *Cell Death Dis*. 2020;11(7):603.
26. Zhou W, Zhou Y, Chen X, Ning T, Chen H, Guo Q, et al. Pancreatic cancer-targeting exosomes for enhancing immunotherapy and reprogramming tumor microenvironment. *Biomaterials*. 2021;268:120546.
27. Tsai HI, Wu Y, Liu X, Xu Z, Liu L, Wang C, et al. Engineered small extracellular vesicles as a FGL1/PD-L1 Dual-Targeting delivery system for alleviating immune rejection. *Adv Sci (Weinh)*. 2022;9(3):e2102634.
28. Shen J, Hu Y, Putt KS, Singhal S, Han H, Visscher DW, et al. Assessment of folate receptor alpha and beta expression in selection of lung and pancreatic cancer patients for receptor targeted therapies. *Oncotarget*. 2018;9(4):4485–95.
29. Yu M, Gai C, Li Z, Ding D, Zheng J, Zhang W, et al. Targeted exosome-encapsulated Erastin induced ferroptosis in triple negative breast cancer cells. *Cancer Sci*. 2019;110(10):3173–82.
30. Feng C, Xiong Z, Wang C, Xiao W, Xiao H, Xie K, et al. Folic acid-modified Exosome-PH20 enhances the efficiency of therapy via modulation of the tumor microenvironment and directly inhibits tumor cell metastasis. *Bioact Mater*. 2021;6(4):963–74.
31. Stern R, Jedrzejewski MJ. Hyaluronidases: their genomics, structures, and mechanisms of action. *Chem Rev*. 2006;106(3):818–39.
32. Zhang J, Yuan ZF, Wang Y, Chen WH, Luo GF, Cheng SX, et al. Multifunctional envelope-type mesoporous silica nanoparticles for tumor-triggered targeting drug delivery. *J Am Chem Soc*. 2013;135(13):5068–73.
33. Wu D, Zhao X, Xie J, Yuan R, Li Y, Yang Q et al. Physical modulation of mesenchymal stem cell exosomes: A new perspective for regenerative medicine. *Cell Prolif*. 2024;57(8):e13630.
34. Harting MT, Srivastava AK, Zhaorigetu S, Bair H, Prabhakara KS, Toledano Furman NE, et al. Inflammation-Stimulated mesenchymal stromal Cell-Derived extracellular vesicles attenuate inflammation. *Stem Cells*. 2018;36(1):79–90.
35. Kim HY, Kim TJ, Kang L, Kim YJ, Kang MK, Kim J, et al. Mesenchymal stem cell-derived magnetic extracellular nanovesicles for targeting and treatment of ischemic stroke. *Biomaterials*. 2020;243:119942.
36. Yang Y, Hong Y, Nam GH, Chung JH, Koh E, Kim IS. Virus-Mimetic fusogenic exosomes for direct delivery of integral membrane proteins to target cell membranes. *Adv Mater*. 2017;29(13).
37. Jia G, Han Y, An Y, Ding Y, He C, Wang X, et al. NRP-1 targeted and cargo-loaded exosomes facilitate simultaneous imaging and therapy of glioma in vitro and in vivo. *Biomaterials*. 2018;178:302–16.
38. Pascucci L, Coccè V, Bonomi A, Ami D, Ceccarelli P, Cusani E, et al. Paclitaxel is incorporated by mesenchymal stromal cells and released in exosomes that inhibit in vitro tumor growth: a new approach for drug delivery. *J Control Release*. 2014;192:262–70.
39. Rayamajhi S. Engineering synthetic and natural vesicular system for tumor-targeted drug delivery. Kansas State University. 2021.
40. Fuhrmann G, Serio A, Mazo M, Nair R, Stevens MM. Active loading into extracellular vesicles significantly improves the cellular uptake and photodynamic effect of porphyrins. *J Control Release*. 2015;205:35–44.
41. Vandergriff A, Huang K, Shen D, Hu S, Hensley MT, Caranasos TG, et al. Targeting regenerative exosomes to myocardial infarction using cardiac homing peptide. *Theranostics*. 2018;8(7):1869–78.
42. Liang L, Zhao L, Wang Y, Wang Y. Treatment for hepatocellular carcinoma is enhanced when Norcantharidin is encapsulated in exosomes derived from bone marrow mesenchymal stem cells. *Mol Pharm*. 2021;18(3):1003–13.
43. Sadeghi S, Tehrani FR, Tahmasebi S, Shafiee A, Hashemi SM. Exosome engineering in cell therapy and drug delivery. *Inflammopharmacology*. 2023;31(1):145–69.
44. Zhuang Z, Zhang Y, Yang X, Yu T, Zhang Y, Sun K, et al. Matrix stiffness regulates the Immunomodulatory effects of mesenchymal stem cells on macrophages via AP1/TSG-6 signaling pathways. *Acta Biomater*. 2022;149:69–81.
45. Kooijmans SAA, Fliervoet LAL, van der Meel R, Fens M, Heijnen HFG, van Bergen PMP et al. PEGylated and targeted extracellular vesicles display enhanced cell specificity and circulation time. *J Control Release*. 2016;224:77–85.
46. Li W, Liu Y, Zhang P, Tang Y, Zhou M, Jiang W, et al. Tissue-Engineered bone immobilized with human adipose stem Cells-Derived exosomes promotes bone regeneration. *ACS Appl Mater Interfaces*. 2018;10(6):5240–54.
47. Xu C, Kang Y, Dong X, Jiang D, Qi M. Integration exosomes with MOF-modified multifunctional scaffold for accelerating vascularized bone regeneration. *Chin Chem Lett*. 2023;34(2):107528.
48. Shi Y, Wang S, Wang K, Yang R, Liu D, Liao H, et al. Relieving macrophage dysfunction by inhibiting SREBP2 activity: A hypoxic mesenchymal stem

- Cells-Derived exosomes loaded multifunctional hydrogel for accelerated diabetic wound healing. *Small*. 2024;20(25):e2309276.
49. Song F, Xu D, Che J, Huang M, Li H. Chitosan hydrogel incorporated with bone marrow mesenchymal stem cell-derived Exosomal TIMP2 to inhibit angiogenesis in cholangiocarcinoma. *Tissue Cell*. 2025;93:102694.
50. Wu D, Tao S, Zhu L, Zhao C, Xu N. Chitosan hydrogel dressing loaded with adipose mesenchymal stem Cell-Derived exosomes promotes skin Full-Thickness wound repair. *ACS Appl Bio Mater*. 2024;7(2):1125–34.
51. Rezaie Z, Ardeshtyrajimi A, Ashkezari MD. Improved anticancer properties of stem cells derived exosomes by prolonged release from PCL nanofibrous structure. *Gene*. 2018;665:105–10.
52. Lashkari M, Rahmani M, Yousefpour Y, Ahmadi-Zeidabadi M, Faridi-Majidi R, Ameri Z, et al. Cell-based wound dressing: bilayered pcl/gelatin nanofibers-alginate/collagen hydrogel scaffold loaded with mesenchymal stem cells. *Int J Biol Macromol*. 2023;239:124099.
53. Chahsetareh H, Yazdian F, Pezeshki-Modaress M, Aleemardani M, Hassanzadeh S, Najafi R, et al. Alginate hydrogel-PCL/gelatin nanofibers composite scaffold containing mesenchymal stem cells-derived exosomes sustain release for regeneration of tympanic membrane perforation. *Int J Biol Macromol*. 2024;262:130141.
54. Wu Y, Liang T, Hu Y, Jiang S, Luo Y, Liu C, et al. 3D Bioprinting of integral ADSCs-NO hydrogel scaffolds to promote severe burn wound healing. *Regen Biomater*. 2021;8(3):rba014.
55. Lu W, Zeng M, Liu W, Ma T, Fan X, Li H, et al. Human urine-derived stem cell exosomes delivered via injectable GelMA templated hydrogel accelerate bone regeneration. *Mater Today Bio*. 2023;19:100569.
56. Hao Z, Ren L, Zhang Z, Yang Z, Wu S, Liu G, et al. A multifunctional neuro-modulation platform utilizing Schwann cell-derived exosomes orchestrates bone microenvironment via immunomodulation, angiogenesis and osteogenesis. *Bioact Mater*. 2023;23:206–22.
57. Hosseini Shamili F, Alibolandi M, Rafatpanah H, Abnous K, Mahmoudi M, Kalantari M, et al. Immunomodulatory properties of MSC-derived exosomes armed with high affinity aptamer toward Myelin as a platform for reducing multiple sclerosis clinical score. *J Control Release*. 2019;299:149–64.
58. Gomari H, Forouzandeh Moghadam M, Soleimani M, Ghavami M, Khodashenas S. Targeted delivery of doxorubicin to HER2 positive tumor models. *Int J Nanomed*. 2019;14:5679–90.
59. Wang C, Wang M, Xu T, Zhang X, Lin C, Gao W, et al. Engineering bioactive Self-Healing antibacterial exosomes hydrogel for promoting chronic diabetic wound healing and complete skin regeneration. *Theranostics*. 2019;9(1):65–76.
60. Li Q, Gong S, Yao W, Yang Z, Wang R, Yu Z, et al. Exosome loaded Genipin crosslinked hydrogel facilitates full thickness cutaneous wound healing in rat animal model. *Drug Deliv*. 2021;28(1):884–93.
61. Zhang M, Hu S, Liu L, Dang P, Liu Y, Sun Z, et al. Engineered exosomes from different sources for cancer-targeted therapy. *Signal Transduct Target Therapy*. 2023;8(1):124.
62. Bao C, Xiang H, Chen Q, Zhao Y, Gao Q, Huang F, et al. A review of labeling approaches used in small extracellular vesicles tracing and imaging. *Int J Nanomed*. 2023;18:4567–88.
63. Gonçalves MS. Fluorescent labeling of biomolecules with organic probes. *Chem Rev*. 2009;109(1):190–212.
64. Zomer A, Maynard C, Verweij FJ, Kamermans A, Schäfer R, Beerling E, et al. In vivo imaging reveals extracellular vesicle-mediated phenocopying of metastatic behavior. *Cell*. 2015;161(5):1046–57.
65. Lai CP, Mardini O, Ericsson M, Prabhakar S, Maguire C, Chen JW, et al. Dynamic biodistribution of extracellular vesicles in vivo using a multimodal imaging reporter. *ACS Nano*. 2014;8(1):483–94.
66. Morishita M, Takahashi Y, Matsumoto A, Nishikawa M, Takakura Y. Exosome-based tumor antigens-adjuvant co-delivery utilizing genetically engineered tumor cell-derived exosomes with immunostimulatory CpG DNA. *Biomaterials*. 2016;111:55–65.
67. Besse B, Charrier M, Lapierre V, Dansin E, Lantz O, Planchard D, et al. Dendritic cell-derived exosomes as maintenance immunotherapy after first line chemotherapy in NSCLC. *Oncotarget*. 2016;5(4):e1071008.
68. Bose RJ, Uday Kumar S, Zeng Y, Afjei R, Robinson E, Lau K, et al. Tumor Cell-Derived extracellular Vesicle-Coated nanocarriers: an efficient theranostic platform for the Cancer-Specific delivery of anti-miR-21 and imaging agents. *ACS Nano*. 2018;12(11):10817–32.
69. Cao Y, Wu T, Zhang K, Meng X, Dai W, Wang D, et al. Engineered Exosome-Mediated Near-Infrared-II region V(2)C quantum Dot delivery for Nucleus-Target Low-Temperature photothermal therapy. *ACS Nano*. 2019;13(2):1499–510.
70. Ahmadi M, Mahmoodi M, Shooran M, Nazari-Khanamiri F, Rezaie J. Harnessing normal and engineered mesenchymal stem cells derived exosomes for cancer therapy: opportunity and challenges. *Int J Mol Sci*. 2022;23(22):13974.
71. Song Y, Song Q, Hu D, Sun B, Gao M, Liang X, et al. The potential applications of artificially modified exosomes derived from mesenchymal stem cells in tumor therapy. *Front Oncol*. 2024;13:1299384.
72. Sheykhasan M, Kalhor N, Sheikholeslami A, Dolati M, Amini E, Fazaeli H. Exosomes of mesenchymal stem cells as a proper vehicle for transfecting miR-145 into the breast cancer cell line and its effect on metastasis. *Biomed Res Int*. 2021;2021(1):5516078.
73. Zhou Y, Dong Y, Zhang A, Wu J, Sun Q. The role of mesenchymal stem cells derived exosomes as a novel nanobiotechnology target in the diagnosis and treatment of cancer. *Front Bioeng Biotechnol*. 2023;11:1214190.
74. Schae D, McBride WH. Opportunities and challenges of radiotherapy for treating cancer. *Nat Reviews Clin Oncol*. 2015;12(9):527–40.
75. Barton MB, Jacob S, Shafiq J, Wong K, Thompson SR, Hanna TP, et al. Estimating the demand for radiotherapy from the evidence: a review of changes from 2003 to 2012. *Radiother Oncol*. 2014;112(1):140–4.
76. de Araujo Farias V, O'Valle F, Serrano-Saenz S, Anderson P, Andrés E, López-Peñalver J, et al. Exosomes derived from mesenchymal stem cells enhance radiotherapy-induced cell death in tumor and metastatic tumor foci. *Mol Cancer*. 2018;17:1–12.
77. Xunian Z, Kalluri R. Biology and therapeutic potential of mesenchymal stem cell-derived exosomes. *Cancer Sci*. 2020;111(9):3100–10.
78. Wu J, Qu Z, Fei ZW, Wu JH, Jiang CP. Role of stem cell-derived exosomes in cancer. *Oncol Lett*. 2017;13(5):2855–66.
79. Lin Z, Wu Y, Xu Y, Li G, Li Z, Liu T. Mesenchymal stem cell-derived exosomes in cancer therapy resistance: recent advances and therapeutic potential. *Mol Cancer*. 2022;21(1):179.
80. Yang F, Wang M, Guan X. Exosomes and mimics as novel delivery platform for cancer therapy. *Front Pharmacol*. 2022;13:1001417.
81. Kalluri R, LeBleu VS. The biology, function, and biomedical applications of exosomes. *Science*. 2020;367(6478):eaa06977.
82. Fan B, Yang S, Wang Y-y, Zhang C, Yang J-p, Wang L-q, et al. Indocyanine green-loaded exosomes for image-guided glioma nano-therapy. *J Exp Nanosci*. 2022;17(1):187–96.
83. Li M, Yin S, Xu A, Kang L, Ma Z, Liu F, et al. Synergistic Phototherapy-Molecular targeted therapy combined with tumor exosome nanoparticles for oral squamous cell carcinoma treatment. *Pharmaceutics*. 2023;16(1):33.
84. Yeo RWY, Lai RC, Zhang B, Tan SS, Yin Y, Teh BJ, et al. Mesenchymal stem cell: an efficient mass producer of exosomes for drug delivery. *Adv Drug Deliv Rev*. 2013;65(3):336–41.
85. Nguyen Cao TG, Kang JH, You JY, Kang HC, Rhee WJ, Ko YT, et al. Safe and targeted sonodynamic cancer therapy using biocompatible exosome-based nanosonosensitizers. *ACS Appl Mater Interfaces*. 2021;13(22):25575–88.
86. Liu Y, Bai L, Guo K, Jia Y, Zhang K, Liu Q, et al. Focused ultrasound-augmented targeting delivery of nanosonosensitizers from homogenous exosomes for enhanced sonodynamic cancer therapy. *Theranostics*. 2019;9(18):5261.
87. Khatami SH, Karami N, Taheri-Anganeh M, Taghviimi S, Tondro G, Khorsand M, et al. Exosomes: promising delivery tools for overcoming blood-brain barrier and glioblastoma therapy. *Mol Neurobiol*. 2023;60(8):4659–78.
88. Sohrabi B, Dayeri B, Zahedi E, Khoshbakht S, Nezamabadi Pour N, Ranjbar H, et al. Mesenchymal stem cell (MSC)-derived exosomes as novel vehicles for delivery of MiRNAs in cancer therapy. *Cancer Gene Ther*. 2022;29(8):1105–16.
89. Dalmizrak A, Dalmizrak O. Mesenchymal stem cell-derived exosomes as new tools for delivery of MiRNAs in the treatment of cancer. *Front Bioeng Biotechnol*. 2022;10:956563.
90. Shan C, Liang Y, Wang K, Li P. Mesenchymal stem cell-derived extracellular vesicles in Cancer therapy resistance: from biology to clinical opportunity. *Int J Biol Sci*. 2024;20(1):347.
91. Vakhshiteh F, Atyabi F, Ostad SN. Mesenchymal stem cell exosomes: a two-edged sword in cancer therapy. *Int J Nanomed*. 2019;14:2847–59.
92. Tan F, Li X, Wang Z, Li J, Shahzad K, Zheng J. Clinical applications of stem cell-derived exosomes. *Signal Transduct Target Therapy*. 2024;9(1):17.
93. Harrell CR, Jovicic N, Djonov V, Volarevic V. Therapeutic use of mesenchymal stem cell-derived exosomes: from basic science to clinics. *Pharmaceutics*. 2020;12(5):474.
94. Janockova J, Slovinska L, Harvanova D, Spakova T, Rosocha J. New therapeutic approaches of mesenchymal stem cells-derived exosomes. *J Biomed Sci*. 2021;28:1–26.

95. Nerkar AG, Banginwar Y, Dawande A. Evolution and applications of CRISPR-Cas9 technology in last decade. *Curr Trends Pharm Pharm Chem*. 2024;6(3):68–72.
96. Xiao-Jie GEL, Li-Juan J, Torres-Ruiz R, Rodriguez-Perales S. CRISPR-Cas9 technology: applications and human disease modelling. *Brief Funct Genomics*. 2017;16(1):4–12.
97. Song Y, Liang F, Tian W, Rayhill E, Ye L, Tian X. Optimizing therapeutic outcomes: preconditioning strategies for MSC-derived extracellular vesicles. *Front Pharmacol*. 2025;16:1509418.
98. Chen S, Sun F, Qian H, Xu W, Jiang J. Preconditioning and engineering strategies for improving the efficacy of mesenchymal stem Cell-Derived exosomes in Cell-Free therapy. *Stem Cells Int*. 2022;2022(1):1779346.
99. Lee B-C, Kang I, Yu K-R. Therapeutic features and updated clinical trials of mesenchymal stem cell (MSC)-derived exosomes. *J Clin Med*. 2021;10(4):711.
100. Prithiviraj S, Garcia AG, Linderfalk K, Yiguang B, Ferveur S, Falck LN, et al. Compositional editing of extracellular matrices by CRISPR/Cas9 engineering of human mesenchymal stem cell lines. *eLife*. 2025;13:RP96941.
101. Duan L, Ouyang K, Xu X, Xu L, Wen C, Zhou X, et al. Nanoparticle delivery of CRISPR/Cas9 for genome editing. *Front Genet*. 2021;12:673286.
102. Rao D, Huang D, Sang C, Zhong T, Zhang Z, Tang Z. Advances in mesenchymal stem cell-derived exosomes as drug delivery vehicles. *Front Bioeng Biotechnol*. 2022;9:797359.
103. Lu Y, Godbout K, Lamothe G, Tremblay JP. CRISPR-Cas9 delivery strategies with engineered extracellular vesicles. *Mol Therapy-Nucleic Acids*. 2023;34.
104. Smyth T, Petrova K, Payton NM, Persaud I, Redzic JS, Graner MW, et al. Surface functionalization of exosomes using click chemistry. *Bioconjug Chem*. 2014;25(10):1777–84.
105. Mukerjee N, Sonar S. Signature of click chemistry in exosome modification for cancer therapeutic. *Clin Translational Discovery*. 2024;4(4):e335.
106. Yang Z, Li Y, Wang Z. Recent advances in the application of mesenchymal stem cell-derived exosomes for cardiovascular and neurodegenerative disease therapies. *Pharmaceutics*. 2022;14(3):618.
107. Matsuzaka Y, Yashiro R. Current strategies and therapeutic applications of mesenchymal stem cell-based drug delivery. *Pharmaceutics*. 2024;17(6):707.
108. Tello RP, Lamparelli EP, Ciardulli MC, Hirvonen J, Barreto G, Mafulli N, et al. Hybrid lipid nanoparticles derived from human mesenchymal stem cell extracellular vesicles by microfluidic sonication for collagen I mRNA delivery to human tendon progenitor stem cells. *Biomaterials Sci*. 2025;13(8):2066–81.
109. Lee J, Lee J-H, Chakraborty K, Hwang J, Lee Y-K. Exosome-based drug delivery systems and their therapeutic applications. *RSC Adv*. 2022;12(29):18475–92.
110. Santos P, Almeida F. Exosome-based vaccines: history, current state, and clinical trials. *Front Immunol*. 2021;12:711565.
111. Mendt M, Kamekar S, Sugimoto H, McAndrews KM, Wu C-C, Gagea M et al. Generation and testing of clinical-grade exosomes for pancreatic cancer. *JCI Insight*. 2018;3(8).
112. Jafarpour S, Ahmadi S, Mokarian F, Sharifi M, Ghobakhloo S, Yazdi M, et al. MSC-derived exosomes enhance the anticancer activity of drugs in 3D spheroid of breast cancer cells. *J Drug Deliv Sci Technol*. 2024;92:105375.
113. Yu S, Liao R, Bai L, Guo M, Zhang Y, Zhang Y, et al. Anticancer effect of hUC-MSC-derived exosome-mediated delivery of PMO-miR-146b-5p in colorectal cancer. *Drug Delivery Translational Res*. 2024;14(5):1352–69.
114. Liu J, Feng Y, Zeng X, He M, Gong Y, Liu Y. Extracellular vesicles-encapsulated let-7i shed from bone mesenchymal stem cells suppress lung cancer via KDM3A/DCLK1/FXYD3 axis. *J Cell Mol Med*. 2021;25(4):1911–26.
115. Gholami A. Cancer stem cell-derived exosomes in CD8+ T cell exhaustion. *Int Immunopharmacol*. 2024;137:112509.
116. Yang J, Teng Y. Harnessing cancer stem cell-derived exosomes to improve cancer therapy. *J Experimental Clin Cancer Res*. 2023;42(1):131.
117. Sonar S, Nyahatkar S, Kalele K. Tumor and cancer stem cells-derived exosomes interplay: A secret of cancer complications. *Clin Translational Discovery*. 2024;4(4):e325.
118. Li X, Li X, Zhang B, He B. The role of cancer stem cell-derived exosomes in cancer progression. *Stem Cells Int*. 2022;2022(1):9133658.
119. Crescitelli R, Lässer C, Lötvall J. Isolation and characterization of extracellular vesicle subpopulations from tissues. *Nat Protoc*. 2021;16(3):1548–80.
120. Luo T, von der Ohe J, Hass R. MSC-derived extracellular vesicles in tumors and therapy. *Cancers*. 2021;13(20):5212.
121. Yang D, Zhang W, Zhang H, Zhang F, Chen L, Ma L, et al. Progress, opportunity, and perspective on exosome isolation-efforts for efficient exosome-based theranostics. *Theranostics*. 2020;10(8):3684.
122. Ibsen SD, Wright J, Lewis JM, Kim S, Ko S-Y, Ong J, et al. Rapid isolation and detection of exosomes and associated biomarkers from plasma. *ACS Nano*. 2017;11(7):6641–51.
123. Lu M, Huang Y. Bioinspired exosome-like therapeutics and delivery nanoplateforms. *Biomaterials*. 2020;242:119925.
124. Li Y-J, Wu J-Y, Liu J, Xu W, Qiu X, Huang S, et al. Artificial exosomes for translational nanomedicine. *J Nanobiotechnol*. 2021;19:1–20.
125. Li Z, Zhou X, Gao X, Bai D, Dong Y, Sun W, et al. Fusion protein engineered exosomes for targeted degradation of specific RNAs in lysosomes: a proof-of-concept study. *J Extracell Vesicles*. 2020;9(1):1816710.
126. Alvarez-Erviti L, Seow Y, Yin H, Betts C, Lakhali S, Wood MJ. Delivery of siRNA to the mouse brain by systemic injection of targeted exosomes. *Nat Biotechnol*. 2011;29(4):341–5.
127. Ohno S-i, Takanashi M, Sudo K, Ueda S, Ishikawa A, Matsuyama N, et al. Systemically injected exosomes targeted to EGFR deliver antitumor MicroRNA to breast cancer cells. *Mol Ther*. 2013;21(1):185–91.
128. Liang Y, Duan L, Lu J, Xia J. Engineering exosomes for targeted drug delivery. *Theranostics*. 2021;11(7):3183.
129. Whiteside TL, editor. Editor exosome and mesenchymal stem cell cross-talk in the tumor microenvironment. *Seminars in immunology*. Elsevier 2018.
130. Wang N, Pei B, Yuan X, Yi C, Wiredu Ocansey DK, Qian H, et al. Emerging roles of mesenchymal stem cell-derived exosomes in Gastrointestinal cancers. *Front Bioeng Biotechnol*. 2022;10:1019459.
131. Jahangiri B, Khalaj-Kondori M, Asadollahi E, Kian Saei A, Sadeghizadeh M. Dual impacts of mesenchymal stem cell-derived exosomes on cancer cells: unravelling complex interactions. *J Cell Communication Signal*. 2023;17(4):1229–47.
132. Lee J-K, Park S-R, Jung B-K, Jeon Y-K, Lee Y-S, Kim M-K, et al. Exosomes derived from mesenchymal stem cells suppress angiogenesis by down-regulating VEGF expression in breast cancer cells. *PLoS ONE*. 2013;8(12):e84256.
133. Biswas S, Mandal G, Roy Chowdhury S, Purohit S, Payne KK, Anadon C, et al. Exosomes produced by mesenchymal stem cells drive differentiation of myeloid cells into immunosuppressive M2-polarized macrophages in breast cancer. *J Immunol*. 2019;203(12):3447–60.
134. Yassine S, Alaeddine N. Mesenchymal stem cell exosomes and cancer: controversies and prospects. *Adv Biology*. 2022;6(2):2101050.
135. O'Brien KP, Khan S, Gilligan KE, Zafar H, Lalor P, Glynn C, et al. Employing mesenchymal stem cells to support tumor-targeted delivery of extracellular vesicle (EV)-encapsulated microRNA-379. *Oncogene*. 2018;37(16):2137–49.
136. Baldari S, Di Rocco G, Magenta A, Picozza M, Toietta G. Extracellular Vesicles-Encapsulated MicroRNA-125b produced in genetically modified mesenchymal stromal cells inhibits hepatocellular carcinoma cell proliferation. *Cells*. 2019;8(12).
137. Xu S, Liu B, Fan J, Xue C, Lu Y, Li C, et al. Engineered mesenchymal stem cell-derived exosomes with high CXCR4 levels for targeted siRNA gene therapy against cancer. *Nanoscale*. 2022;14(11):4098–113.
138. Bliss SA, Sinha G, Sandiford OA, Williams LM, Engelberth DJ, Guirio K, et al. Mesenchymal stem Cell-Derived exosomes stimulate cycling quiescence and early breast Cancer dormancy in bone marrow. *Cancer Res*. 2016;76(19):5832–44.
139. Zhang Z, Guo X, Guo X, Yu R, Qian M, Wang S, et al. MicroRNA-29a-3p delivery via exosomes derived from engineered human mesenchymal stem cells exerts tumour suppressive effects by inhibiting migration and vasculogenic mimicry in glioma. *Aging*. 2021;13(4):5055–68.
140. Luo T, Liu Q, Tan A, Duan L, Jia Y, Nong L, et al. Mesenchymal stem Cell-Secreted exosome promotes chemoresistance in breast Cancer via enhancing miR-21-5p-Mediated S100A6 expression. *Mol Ther Oncolytics*. 2020;19:283–93.
141. Lyu T, Wang Y, Li D, Yang H, Qin B, Zhang W, et al. Exosomes from BM-MSCs promote acute myeloid leukemia cell proliferation, invasion and chemoresistance via upregulation of S100A4. *Exp Hematol Oncol*. 2021;10(1):24.
142. Jia Z, Zhu H, Sun H, Hua Y, Zhang G, Jiang J, et al. Adipose mesenchymal stem Cell-Derived Exosomal microRNA-1236 reduces resistance of breast Cancer cells to cisplatin by suppressing SLC9A1 and the Wnt/ β -Catenin signaling. *Cancer Manag Res*. 2020;12:8733–44.
143. Dong B, Li S, Zhu S, Yi M, Luo S, Wu K. MiRNA-mediated EMT and CSCs in cancer chemoresistance. *Exp Hematol Oncol*. 2021;10(1):12.
144. Vallabhaneni KC, Penfornis P, Dhule S, Guillonau F, Adams KV, Mo YY, et al. Extracellular vesicles from bone marrow mesenchymal stem/stromal cells transport tumor regulatory microRNA, proteins, and metabolites. *Oncotarget*. 2015;6(7):4953–67.

145. Kanchanapally R, Khan MA, Deshmukh SK, Srivastava SK, Khushman M, Singh S, et al. Exosomal formulation escalates cellular uptake of Honokiol leading to the enhancement of its antitumor efficacy. *ACS Omega*. 2020;5(36):23299–307.
146. Zhou Y, Zhou W, Chen X, Wang Q, Li C, Chen Q, et al. Bone marrow mesenchymal stem cells-derived exosomes for penetrating and targeted chemotherapy of pancreatic cancer. *Acta Pharm Sin B*. 2020;10(8):1563–75.
147. Rajendran RL, Paudel S, Gangadaran P, Oh JM, Oh EJ, Hong CM et al. Extracellular vesicles act as Nano-Transporters of tyrosine kinase inhibitors to revert iodine avidity in thyroid Cancer. *Pharmaceutics*. 2021;13(2).
148. Sun H, Zhu R, Guo X, Zhao P, Zhang R, Zhao Z, et al. Exosome miR-101–3p derived from bone marrow mesenchymal stem cells promotes radiotherapy sensitivity in non-small cell lung cancer by regulating DNA damage repair and autophagy levels through EZH2. *Pathol - Res Pract*. 2024;256:155271.
149. Gao Y, Liu J, Wu M, Zhang Y, Wang M, Lyu Q, et al. Photosensitive hybrid $\gamma\delta$ -T exosomes for targeted Cancer photoimmunotherapy. *ACS Nano*. 2025;19(4):4251–68.
150. Yang J, Wang Q, Xing T, Wang X, Li G, Shang Z, et al. Engineered exosome-mediated Cobalt sulfide quantum Dot targeted delivery for photothermal and chemodynamic anticancer therapy. *J Drug Deliv Sci Technol*. 2023;83:104441.
151. Huang L, Xu C, Xu P, Qin Y, Chen M, Feng Q, et al. Intelligent photosensitive mesenchymal stem cells and Cell-Derived microvesicles for photothermal therapy of prostate Cancer. *Nanotheranostics*. 2019;3(1):41–53.
152. Sancho-Albero M, Encinas-Giménez M, Sebastián V, Pérez E, Luján L, Santa-maría J, et al. Transfer of photothermal nanoparticles using stem cell derived small extracellular vesicles for in vivo treatment of primary and multinodular tumours. *J Extracell Vesicles*. 2022;11(3):e12193.
153. Whiteside TL. Tumor-Derived exosomes and their role in Cancer progression. *Adv Clin Chem*. 2016;74:103–41.
154. Tian Y, Li S, Song J, Ji T, Zhu M, Anderson GJ, et al. A doxorubicin delivery platform using engineered natural membrane vesicle exosomes for targeted tumor therapy. *Biomaterials*. 2014;35(7):2383–90.
155. Hanahan D, Weinberg RA. Hallmarks of cancer: the next generation. *Cell*. 2011;144(5):646–74.
156. Phinney DG, Pittenger MF. Concise review: MSC-Derived exosomes for Cell-Free therapy. *Stem Cells*. 2017;35(4):851–8.
157. Baglio SR, Pegtel DM, Baldini N. Mesenchymal stem cell secreted vesicles provide novel opportunities in (stem) cell-free therapy. *Front Physiol*. 2012;3:359.
158. Melo SA, Luecke LB, Kahlert C, Fernandez AF, Gammon ST, Kaye J, et al. Glypican-1 identifies cancer exosomes and detects early pancreatic cancer. *Nature*. 2015;523(7559):177–82.
159. Lener T, Gimona M, Aigner L, Börger V, Buzas E, Camussi G, et al. Applying extracellular vesicles based therapeutics in clinical trials - an ISEV position paper. *J Extracell Vesicles*. 2015;4:30087.

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.